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HINTS & SOLUTIONS

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CHEMISTRY

Class – 9

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1. Structure of The Atom

Solutions

Worksheet –1

1. **Correct Answer: A) Democritus**
   Explanation: Democritus, a Greek philosopher (around 400 BCE), proposed that matter is composed of indivisible particles called atoms.
2. **Correct Answer: D) Maharshi Kanada**
   Explanation: In ancient India, Maharshi Kanada proposed the concept of "Anu" (atom) as the smallest indivisible particle of matter. The word atom comes from the Greek word "atomos," meaning indivisible, but the concept was also independently proposed by Kanada.
3. **Correct Answer: A) Atom**
   Explanation: Elements are pure substances made of only one type of atom.
4. **Correct Answer: D) Maharshi Kanada**
   Explanation: Maharshi Kanada in ancient Indian philosophy introduced Paramanu as the smallest particle of matter, similar to the modern concept of an atom.
5. **Correct Answer: B) -1.6×10^{-19} C**
   Explanation: The electron carries a negative charge of 1.6×10^{-19} C. The negative sign indicates its charge is negative.
6. **Correct answer:A) Goldstein**
   Explanation: Anode rays (positive rays) were discovered by Eugen Goldstein in 1886 while working with cathode ray tubes.
7. **Correct Answer: B) It is an ionised hydrogen atom.**
   Explanation: A proton is essentially a hydrogen atom stripped of its electron, leaving a positively charged particle.
8. **Correct Answer: B) 1.76×10^{11} C/kg**
   Explanation: J. J. Thomson measured the charge-to-mass ratio (e/m) of an electron and found it to be 1.76×10^{11} C/kg.
9. **Correct Answer: B) Chadwick**
   Explanation: James Chadwick discovered the neutron in 1932, showing that atoms also have neutral particles in the nucleus.
10. **Correct Answer: C) Slightly more than that of a proton.**
    Explanation: The neutron's mass is slightly larger than a proton:
    - Proton mass ≈ 1.6726×10^{-27} kg
    - Neutron mass ≈ 1.6750×10^{-27} kg

Worksheet-2

1. **Correct Answer: A) Sphere in shape**
   Explanation: Thomson proposed the "plum pudding model" in 1904, where the atom is a positively charged sphere with electrons embedded like "plums in pudding
2. **Correct Answer: B) α-rays scattering experiment**
   Explanation: Rutherford's nuclear model (1911) came from the gold foil experiment, where α-particles were scattered, showing a tiny, dense, positively charged nucleus.
3. **Correct Answer: C) $nh/2\pi$**
   Explanation: According to Bohr's postulate, the angular momentum of an electron in orbit is quantized:
4. **Correct Answer: D) N, O-shells**
   Explanation: Energy difference ΔE between orbits decreases with higher n (farther from nucleus). So the difference is smallest for outermost shells, e.g., N → O.

5. **Correct answer:$3h/2\pi$**
   Explanation:
   - M-shell corresponds to n = 3.
   - Using Bohr's formula: $L=nh/2\pi=3h/2\pi$
6. **Correct answer:$r_n=n^2r_1$**
   Explanation: Bohr's radius formula: $r_n=n^2r_1, r_1\approx0.529$ A°
   Where n is the principal quantum number.
7. **Correct Answer: D) Stationary states**
   Explanation: Bohr postulated electrons move in certain allowed orbits without radiating energy, called stationary states.
8. **Correct Answer: A) K, L-shells**
   Explanation: Energy difference is largest for inner orbits, i.e., between K (n=1) and L (n=2), because electron is closer to nucleus → higher binding energy.
9. **Correct Answer: B) $r_n=n^2r_1$**
   Explanation: Radius increases with n^2
10. **Correct Answer: A) n/k**
    Explanation: In Sommerfeld's extension of Bohr's model:
    - Elliptical orbits are characterized by two quantum numbers: n (principal) and k (azimuthal)
    - Eccentricity of orbit = k/n

Worksheet-3

1. **Correct Answer: B) Two**
   Explanation:
   - Aufbau principle: Electrons fill orbitals in order of increasing energy.
   - Each orbital can hold two electrons with opposite spins (same n, l, m_l, but opposite m_s).
2. **Correct answer:$2n^2$**
   Explanation:
   - Each main shell (n) can have a maximum of $2n^2$ electrons.
   - Example: n = 1 → $2\times1^2 = 2$ electrons, n = 2 → 8 electrons, etc.
3. **Correct answer:C) Two**
   Explanation:
   - Each orbital (defined by n, l, m_l) can have 2 electrons with opposite spins (Pauli principle).
4. **Correct Answer: C) Hund's rule**
   Explanation:
   - Hund's rule: Electrons occupy degenerate orbitals (same energy) singly first, with parallel spins, before pairing.
5. **Correct Answer: D) $1s^2\ 2s^2\ 2p^6$**
   Explanation: the perfect sequence of increasing energy: 1s is filled, then 2s is filled, and then 2p is filled (this configuration represents Neon)
6. **Correct Answer: A) Pauli's exclusion principle**
   Explanation:
   Pauli Exclusion Principle (1925): No two electrons in an atom can have the same n,
7. **Correct Answer: B) Stability**
   Explanation:
   - Half-filled (e.g., p^3) or fully filled (e.g., p^6) sub-shells are energetically stable.
   - l, m_l, m_s.
8. **Correct Answer: A) Nitrogen atom (ground state)**
   Explanation:
   - Total electrons = 7 → element = Nitrogen
   - Electrons are distributed in $1s^2\ 2s^2\ 2p^3$
   - Unpaired electrons in 2p orbitals are distributed singly in all three orbitals, following Hund's rule → ground state configuration
9. **Correct Answer: D) Degenerate orbital**
   Explanation:
   - Degenerate orbitals = orbitals of same energy in a sub-shell.
   - Hund's rule says: Fill singly with parallel spins before pairing.
10. **Correct Answer: A) Electronic configuration**
    Explanation:
    - Electronic configuration: Shows distribution of electrons in shells/sub-shells following Aufbau, Pauli, and Hund rules.

COMPETETIVE CORNER

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: B) 9.58×10^4 C/g**
   Explanation: The specific charge (e/m ratio) of a particle is its charge divided by its mass.
   - Charge of a proton (e): 1.602×10^{-19} C.
   - Mass of a proton (m): 1.672×10^{-24} g.
   - Calculation: $e/m = (1.602 \times 10^{-19}\ C) / (1.672 \times 10^{-24}\ g) \approx 9.58 \times 10^4\ C/g$
2. **Correct Answer: D) 0**
   Explanation: Neutrons are subatomic particles found in the nucleus that are electrically neutral. Unlike protons (+1) and electrons (-1), neutrons carry no net electric charge.
3. **Correct Answer: A) Nature of gas filled in the discharge tube.**
   Explanation**:** Anode rays (canal rays) are composed of positive ions formed when electrons knock other electrons off gas atoms. Because different gases have different atomic masses and can form ions with different charges, the properties (like mass and e/m ratio) of these rays change depending on the gas used.
4. **Correct Answer: D) Si^{+4} (A = 28)**
   Explanation: Specific charge is calculated as Charge / Mass Number (A).
   Si^{+4}: $4 / 28 \approx 0.1429$
   Since 0.1429 is the largest value, Si^{+4} has the highest specific charge
5. **Correct Answer: A) Electrical neutrality**
   Explanation: J.J. Thomson's "Plum Pudding" model proposed that an atom is a sphere of positive charge with electrons embedded in it. This model successfully explained why atoms are electrically neutral (the total positive charge equals the total negative charge), but it could not explain the arrangement of electrons in orbits or the existence of a nucleus.
6. **Correct Answer: C) Nuclei of helium atoms.**
   Explanation: Alpha (α) particles are identical to the nucleus of a helium atom. They consist of two protons and two neutrons, giving them a mass of 4 amu and a charge of +2.
7. **Correct Answer: A) Planetary model**
   Explanation: Rutherford's model is called the Planetary model because it describes the atom as having a central, dense nucleus with electrons "orbiting" around it, much like planets orbit the sun.
8. **Correct Answer: B) Energy is released**
   Explanation: The K shell (n=1) is the lowest energy level, while the N shell (n=4) is higher. When an electron moves from a higher energy level to a lower one, the excess energy is released as radiation.
9. **Correct Answer: B) 3:2**
   Explanation: Radius $r \propto n^2/Z$
   - For Li^{+2} (n=3, Z=3): $r_1 \propto 3^2/3 = 3$.
   - For He^+ (n=2, Z=2): $r_2 \propto 2^2/2 = 2$.
   - Ratio $r_1 : r_2 = 3 : 2$.
10. **Correct Answer: C) $mvr = Kh/2\pi$**
    Explanation: Bohr's second postulate states that the angular momentum (L = mvr) is quantized and is an integral multiple of $h/2\pi$
11. **Correct Answer: B) The electronic valency configuration of an element with atomic number 23 is**
    - $3d^3\ 4s^2$
    - Explanation: Atomic number 23 is Vanadium. Its full configuration is $1s^2\ 2s2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^3$. The valency shell configuration is $3d^3\ 4s^2$.

12. **Correct Answer: C) Excited state of a neon atom**
    Explanation: Total electrons = 2+2+5+1 = 10. Neon normally has 10 electrons in the configuration $1s^2\ 2s^2\ 2p^6$. Here, one electron from 2p has jumped to 3s, indicating an excited state.
13. **Correct Answer: A) nl^x**
    Explanation: This is the standard notation where n is the principal quantum number, l is the subshell (s, p, d, f), and x is the number of electrons in that subshell.
14. **Correct Answer: A) Moeller's diagram**
    Explanation: Moeller's diagram (also known as the Aufbau diagram) provides the "diagonal rule" to determine the filling order of subshells.
15. **Correct Answer: A) $1s^2\ 2s^2\ 2p^3$**
    Explanation: Nitrogen has an atomic number of 7. Electrons are distributed as 2 in 1s, 2 in 2s, and the remaining 3 in 2p.
16. **Correct Answer: B) 4p**
    Explanation: According to the (n+l) rule and the Aufbau principle, the energy level sequence follows: ... 4s → 3d → 4p.
17. **Correct Answer: A) Fe^{2+} (Z = 26)**

    Explanation: Magnetic moment increases with the number of unpaired electrons (n).

    Fe^{2+} ($3d^6$): 4 unpaired electrons.
18. **Correct Answer: B) Ionisation energy of Li^{+2} ion (H-atom = 13.6 eV):**
    - 122.4 eV
    - Explanation: IE = 13.6 x Z^2 eV. For Li^{+2}, Z=3. So, 13.6 x 3^2 = 13.6 x 9 = 122.4 eV.
19. **Correct Answer: C) Energy required to remove both electrons from helium:**
    - 79 eV
    - Explanation: First IE of He is ≈ 24.6 eV. The second IE (removing an electron from He^+) is 13.6 x 2^2 = 54.4 eV. Total energy = 24.6 + 54.4 = 79 eV.
20. **Correct Answer: A) First**
    Explanation: Total Energy (E_n) = 1/2 xPotential Energy. So, E_n = -3.4 eV.
    Using $E_n = -13.6/n^2$, we get $-3.4 = -13.6/n^2 \rightarrow n^2 = 4 \rightarrow n = 2$.
    The state n=2 is the First excited state.
21. **Correct Answer: A) Neon**
    Explanation: K-shell = 2 electron; L-shell = 8 electrons
    Total electrons = 2 + 8 = 10
    Atomic number 10 corresponds to Neon.
22. **Correct Answer: B) Energy is emitted**
    Explanation: Moving from higher orbit (3rd) → lower orbit (1st)
    - When an electron comes closer to the nucleus, it loses energy

    This energy is released as light (emitted)

ONE OR MORE THAN ONE OPTION CORRECT TYPE

23. **Correct Answer: A, B, and D**
    Explanation: Orbitals fill in increasing order of energy (A). Lower (n+l) values fill first (B). If (n+l) is equal, the orbital with lower n fills first (D).
24. **Correct Answer: A, B, and D**
    Explanation: Electrons occupy degenerate orbitals singly first (A) with parallel spins (B) to obtain the maximum number of unpaired electrons (D).
25. **Correct Answer: A, B, and D**
    Explanation: Isobars have the same mass number (the upper number). C-14 and N-14 (A), Ar-40 and Ca-40 (B), and K-40 and Ca-40 (D) all share the same mass numbers.
26. **Correct Answer: A, B, C, and D**
    Explanation: Chromium's ground state is [Ar] $3d^5\ 4s^1$ (A). It has 6 unpaired electrons (B). It is stable due to the half-filled 3d subshell (C) and maximum exchange energy (D).
27. **Correct Answer: A, B, and C**
    Explanation:

    The energy of an orbital is determined by the sum of its principal quantum number (n) and its azimuthal quantum number (l). The values for l are: s=0, p=1, d=2, f=3.

ASSERTION AND REASNING TYPE

28. Correct Answer: A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A.

Explanation:

- Isoelectronic species are atoms or ions that have the same total number of electrons.
- Neon (Ne): Atomic number is 10, so it has 10 electrons.
- Sodium ion (Na^+): Atomic number is 11. The +1 charge means it has lost one electron (11 - 1 = 10), so it also has 10 electrons.
- Since they both have 10 electrons, they are isoelectronic, and the reason provided directly defines the term used in the assertion

29. Correct Answer: A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A.

Explanation:

- Thomson's Model: J.J. Thomson proposed that an atom consists of a uniform sphere of positive electricity.
- The Analogy: He compared this to a "plum pudding" or a watermelon, where the positive charge is the pudding/fruit and the electrons are the "plums" or seeds embedded within it.
- The reason correctly describes the structural arrangement that led to the specific nickname "plum pudding model".

30. Correct answer: B) Both A and R are true, but R is NOT the correct explanation of A

Explanation:

- Assertion is True: Ernest Rutherford's gold foil experiment proved that the atom's mass and positive charge are concentrated in a tiny central region called the nucleus.
- Reason is True: During the experiment, the vast majority of alpha particles passed straight through the gold foil without any deflection.
- Why R is not the correct explanation: The fact that most particles passed through *without* deflection explains that the atom is mostly empty space. It was the rare, large-angle deflections and "bounce-backs" (1 in 8,000) that specifically proved the mass was concentrated in a small, dense, positively charged nucleus.

31. Correct Answer: C) A is true, but R is false

Explanation:

- Assertion is True: Niels Bohr proposed the concept of stationary states or discrete energy levels, stating that electrons revolve around the nucleus in fixed orbits without losing energy.
- Reason is False: Electrons do not emit energy when jumping from a lower to a higher level; instead, they absorb energy to make that jump. Energy is emitted (released) only when an electron jumps from a higher energy level (like the N shell) to a lower energy level (like the K shell).

MATCH THE FOLLOWING TYPE

32. Correct answer:

A. Maximum electrons in a shellI) $2n^2$According to the Bohr-Bury scheme, the maximum number of electrons that can be accommodated in a shell is given by the formula $2n^2$, where n is the principal quantum number (shell number).

B. Outermost shell can holdII) 8 electronsThe Octet Rule states that the maximum number of electrons that can be accommodated in the outermost shell is 8, regardless of whether the shell has the capacity to hold more (except for the K-shell).

C. K-shell maximum electronsIII) 2 electronsFor the K-shell, n = 1. Using the $2n^2$ formula: $2(1)^2 = 2$. Thus, the first shell can hold a maximum of 2 electrons.

D. Electrons fill shells from IV) Inner to outer shellsThis follows the Aufbau Principle, which states that electrons are filled in increasing order of their energies. Inner shells (lower energy) are filled before outer shells (higher energy).

33. Correct answer:

Hydrogen (Z = 1): Has only 1 electron → goes in 1s orbital → $1s^1$

Carbon (Z = 6): Electrons fill as 2 in 1s, 2 in 2s, and 2 in 2p → $1s^2\ 2s^2\ 2p^2$

Oxygen (Z = 8): 2 in 1s, 2 in 2s, and 4 in 2p → $1s^2\ 2s^2\ 2p^4$

Sodium (Z = 11): After noble gas Ne (10), next electron goes into 3s → $3s^1$

COMPREHENSION TYPE

I

34. Correct Answer: B) Fixed energy shells

Explanation:

According to Bohr's atomic model, electrons do not move randomly. They revolve around the nucleus in definite circular paths or shells, each having a fixed amount of energy. These shells are also called energy levels.
So, electrons move in fixed energy shells, not inside the nucleus or in straight lines.

35. Correct Answer: C) Maximum electrons in a shell

Explanation:
The $2n^2$ formula gives the maximum number of electrons that can be accommodated in a shell, where n = shell number.
Example:

- K shell (n = 1) → $2(1)^2 = 2$ electrons
- L shell (n = 2) → $2(2)^2 = 8$ electrons

Hence, it is used to find the maximum electrons in a shell.

36. Correct Answer: B) Small and positively charged

Explanation:

According to Rutherford's nuclear model, the nucleus is:

- Very small in size compared to the atom
- Positively charged because it contains protons
- Contains most of the mass of the atom

The rest of the atom is mostly empty space where electrons move around the nucleus.

II

37. Correct Answer: C) Empty space

Explanation:
Rutherford's gold foil experiment showed that most alpha particles passed straight through the foil without deflection. This means that most of the atom is empty space, with a small, dense, positively charged nucleus at the center and electrons moving around it.

38. Correct Answer: C) Eight electrons in the outermost shell

Explanation:
The octet rule states that atoms tend to gain, lose, or share electrons to achieve a stable configuration of eight electrons in their valence (outermost) shell, similar to noble gases.

39. **Correct Answer: C) Bohr**
Explanation:
Niels Bohr proposed that electrons revolve around the nucleus in specific, fixed energy levels (shells) and do not lose energy while moving in these orbits. This model explained the stability of atoms and the line spectrum of hydrogen.

INTEGER TYPE

40. **Correct Answer: Using $2n^2$ rule:**
- K shell → 2 electrons
- L shell → 8 electrons
Remaining = 16 − (2 + 8) = 6 electrons

41. **Correct Answer: Neutrons = Mass number − Protons**
Neutrons = 23 − 11 = 12

42. **Correct Answer: Outermost shell = 3rd shell (n = 3)**
Electrons in this shell:
- $3s^2$ → 2 electrons
- $3p^3$ → 3 electrons
Total = 2 + 3 = 5 electrons

43. **Correct Answer: According to Rutherford's model, the nucleus contains:**
- Protons (positively charged)
- Neutrons (neutral) *(added later by Chadwick, but now* included in nuclear composition)

Number of types of particles in nucleus = 2

Previous year questions

1. **Correct Answer: B)**
Explanation:
Statement I is Correct**:** According to classical electromagnetism, an accelerating charged particle (like an electron orbiting a nucleus) should continuously radiate energy. This would cause the electron to lose energy and eventually spiral into the nucleus, making the atom unstable. Rutherford's model could not explain why this doesn't happen.
Statement II is Incorrect**:** In the electromagnetic spectrum, X-rays have a very high frequency and a very short wavelength. Microwaves, on the other hand, have a much longer wavelength.

2. **Correct Answer: B**
Explanation:
4s: $n = 4, l = 0$, so m_l must be 0.
3p: $n = 3, l = 1$, so m_l can be -1, 0, +1.
5d: $n = 5, l = 2$, so m_l can be -2, -1, 0, +1, +2.

3. **Correct Answer: C)**
Explanation:
Assertion (A) is True**:** Like any neutral atom, isotopes have an equal number of protons and electrons, making them electrically neutral.
Reason (R) is False**:** This is actually the definition of isobars. Isotopes are defined as species with the same atomic number (same number of protons) but different mass numbers (different number of neutrons).

4. **Correct Answer: C)**
Explanation:
II. Proton numbers of elements X and Y are 6 and 7 respectively.
Correct**.** X has 6 protons and Y has 7 protons.
III. Elements X and Y are represented by ${}^{14}_{6}X$ and ${}^{14}_{7}Y$ respectively.
Correct. These symbols correctly match our calculated atomic and mass numbers.
IV. Elements X and Y are isobars.
Correct**.** Isobars are atoms of different elements (different atomic numbers) that have the same mass number (nucleon number). Since both have a mass number of 14, they are isobars.

5. **Correct Answer: C)**
Explanation:
P and Q are isotopes: Correct**.** Isotopes are atoms of the same element (same number of protons) that have different numbers of neutrons (and thus different mass numbers).

6. Correct Answer: B

Explanation:

- Element X: Shows an M-shell (3rd shell) containing 2 electrons. This means its full configuration is 2, 8, 2.
- Valency of X: Since it has 2 valence electrons, it tends to lose them to become stable. Its valency is 2.
- Element Y: Shows an L-shell (2nd shell) containing 6 electrons. This means its configuration is 2, 6.
- Valency of Y: Since it has 6 valence electrons, it needs to gain $8 - 6 = 2$ electrons to complete its octet. Its valency is 2.

7. Correct Answer: C)

Explanation:

Element P: Has 5 electrons in the L-shell. Since the K-shell (inner) must be full with 2 electrons, the total electrons = $2 + 5 = 7$. Thus, the atomic number is 7 (Nitrogen).

Element Q**:** Has 5 electrons in the M-shell. This implies the K and L shells are full (2 + 8). Total electrons = $2 + 8 + 5 = 15$. Thus, the atomic number is 15 (Phosphorus).

Valency: Both have 5 valence electrons, so they gain 3 electrons to reach stability. Their valency is 3, not 5.

8. Correct Answer: D)

Explanation:

Statement III (Incorrect): P has 16 protons (Sulfur) and 18 electrons (S^{2-}). It is an ion, not a neutral noble gas.

Statement IV (Incorrect): Q has 11 protons and 10 electrons (11-10 = +1). Q is a cation, not an anion**.** U is neutral (20p, 20e).

9. Correct answer:C)

Explanation:

Proton Numbers (Z): U(19), V(53), W(17), X(35), Y(18). Order: W(17) < Y(18) < U(19) < X(35) < V(53). (Statement I is incorrect).

Mass Numbers (A): W(35) < U(39) < Y(40) < X(80) < V(127). (Statement II is correct).

Neutron Check: In U, neutrons = 39 - 19 = 20. In W, neutrons = 35 - 17 = 18. Protons do not equal neutrons in all elements. (Statement IV is incorrect).

10. Correct Answer: C)

Explanation:

- R (11): Configuration 2, 8, 1. It loses 1 electron to form R^+ (univalent cation).
- T (17): Configuration 2, 8, 7. It gains 1 electron to form T^- (univalent anion).
- Compound: R and T will combine to form RT (like NaCl), not R_2T.
- P (4): Configuration 2, 2. It has 2 electrons in its L-shell, not a complete octet (8)

2. The Periodic Table

Solutions

Worksheet –1

1. **Correct Answer: B**
   Antoine Lavoisier was the first to broadly classify elements into metals and non-metals.
2. **Correct Answer: A**
   John Newlands proposed the Law of Octaves, stating that every 8th element has properties similar to the 1st.
3. **Correct Answer: D**
   The Law of Octaves works correctly only up to calcium; beyond this, the periodic pattern breaks down.
4. **Correct Answer: A**
   In the atomic volume vs atomic mass curve drawn by Lothar Meyer, alkali metals show the maximum atomic volumes, so they appear at the peaks.
5. **Correct Answer: C**
   Dmitri Mendeleev arranged elements in increasing order of their atomic weights (atomic masses) and grouped them by similar properties.
6. **Correct Answer: A**
   According to John Newlands, counting eight elements from sodium gives: Na(1), Mg(2), Al(3), Si(4), P(5), S(6), Cl(7), P(8).
7. **Correct Answer: A**
   The discovery of inert (noble) gases added a new group to the periodic table, which did not fit into Newlands' octave pattern, proving it incomplete.
8. **Correct Answer: C**
   Atomic weight of Ca ≈ 40 and Ba ≈ 137. Arithmetic mean = (40 + 137) / 2 ≈ 88, which is approximately equal to the atomic weight of strontium (≈ 87.6). This follows the idea used in Dobereiner's triads.
9. **Correct Answer: C**
   Transition metals like Iron (Fe), Cobalt (Co), Nickel (Ni), Copper (Cu), and the Platinum group metals have very small atomic volumes because their atoms are more tightly packed due to strong interatomic forces.
10. **Correct Answer: B**
    Dmitri Mendeleev arranged the elements into 6 horizontal rows (periods) in his periodic table based on increasing atomic weight and similar chemical properties.

Worksheet – 2

1. **Correct Answer: C**
   2nd period = 8 elements, 3rd = 8, 4th = 18, 5th = 18 in the modern periodic table.
2. **Correct Answer: A**
   Modern periodic law states that physical and chemical properties of elements depend on their atomic number, not atomic mass.
3. **Correct Answer: C**
   Elements in the same group have the same valence shell electronic configuration, hence similar chemical properties.
4. **Correct Answer: C**
   Hydrogen has one valence electron ($1s^1$), like alkali metals (ns^1), so it resembles them in electronic configuration.
5. **Correct Answer: C**
   The 6th period is the longest (32 elements) and the 1st period is the shortest (2 elements).
6. **Correct Answer: C**
   The modern (long form) periodic table consists of 7 horizontal rows called periods.

7. **Correct Answer: B**
Each period starts with an alkali metal. The 5th period begins with rubidium (Rb).
8. **Correct Answer: D**
The modern (long form) periodic table has 18 vertical columns called groups.
9. **Correct Answer: B**
Inner transition elements = Number of lanthanides + number of actinides = 14 + 14 = 28
10. **Correct Answer: C**
Atomic number 15 corresponds to phosphorus (P). Phosphorus lies in group 15 (pnictogens) and period 3 of the periodic table.

Worksheet – 3

1. **Correct Answer: B**
The regular repetition of similar physical and chemical properties at definite intervals is known as periodicity.
2. **Correct Answer: C**
Electropositive elements lose electrons easily. Sodium (Na), an alkali metal, loses one electron readily and is more electropositive than Ni, Fe, and Cl.
3. **Correct Answer: D**
Electronegativity increases across a period and is high for halogens. Among the given options, chlorine has the highest electronegativity.
4. **Correct Answer: A.**
Metallic character decreases from left to right across a period. Sodium, being the leftmost element among the options, has the highest metallic character.
5. **Correct Answer: D**
Atomic radius increases down a group due to the addition of new shells. Among Li, Na, K, and Rb, Rb is lowest in Group 1, so it has the largest atomic radius.
6. **Correct Answer: B.**
Metallic radius is measured between two nearest neighbouring metal atoms in a metallic lattice and is taken as half that distance.
7. **Correct Answer: B**
Covalent radius is the smallest (weak attraction), metallic radius is intermediate, and covalent radius is the largest because metal atoms are loosely packed compared to covalent bonding.
8. **Correct Answer: A**
Valency of a non-metal = 8 − number of valence electrons = 8 − 6 = 2.
9. **Correct Answer: A**
Down a group, atomic size increases and valence electrons are lost more easily. Hence, metallic character increases.
10. **Correct Answer: B**
An angstrom is a unit of length used to measure atomic distances.
1 Å = 10^{-10} metre.

Competitive Corner

SINGLE CORRECT OPTION TYPE

1. **Correct Answer: B**
Van der Waals radius refers to the effective size of an atom when it is not bonded, measured as half the distance between two nearest non-bonded atoms of adjacent molecules.
2. **Correct Answer: A**
Alkali metals have the largest atomic volumes in each period, so they appear at the peaks of Lothar Meyer's atomic volume vs atomic mass curve.
3. **Correct Answer: B.**
Horizontal rows in Mendeleev's periodic table are called periods, while vertical columns are called groups.
4. **Correct Answer: B.**
As we move from left to right in a period, the atomic number increases by one for each element, so the nuclear charge (number of protons) increases.
5. **Correct Answer: C.**
The length of a period depends on how many electrons can be accommodated in the available shells/subshells (according to electronic configuration).

6. **Correct Answer: C.**
Across a period, valency increases from 1 to 4 and then decreases to 0; down a group, valency generally remains the same.
7. **Correct Answer: B.**
Across a period, nuclear charge increases, so atoms attract electrons more strongly (electronegativity increases ↑) and lose electrons less easily (electropositivity decreases ↓).
8. **Correct Answer: C.**
Across a period, protons are added to the nucleus (nuclear charge increases) while electrons are added to the same shell. The stronger nuclear attraction pulls the electron cloud closer to the nucleus, so atomic radius decreases.
9. **Correct Answer: C.**
Group 18 elements (noble gases) have completely filled valence shells, making them very stable and largely unreactive. Hence, they exist as gases and show almost no chemical activity under normal conditions.
10. **Correct Answer: B.**
The position of an element in the periodic table provides information about its atomic number, electronic configuration, valency, and number of valence electrons, which helps predict its chemical behavior.
11. **Correct Answer: B.**
Transition elements generally show variable valency and form coloured ions and compounds due to the presence of partially filled d-orbitals. They are also metals with high melting points and good electrical conductivity.
12. **Correct Answer: C**
Inner transition elements are the lanthanides and actinides of the sixth and seventh periods belonging to Group 3.
13. **Correct Answer: C**
Inert gases have weak intermolecular forces, so they possess low melting and boiling points, and due to their stable electronic configuration, they are chemically inert (do not undergo oxidation or combustion).
14. **Correct Answer: B.**
Down the halogen group, the physical state changes regularly: fluorine and chlorine are gases, bromine is a liquid, and iodine is a solid.
15. **Correct Answer: B.**
Alkaline earth metals are greyish-white, harder than alkali metals, and they lose two electrons to form M^{2+} ions, mainly forming ionic compounds (except beryllium compounds).
16. **Correct Answer: C.**
Alkali metals lose one electron easily to form M^{+} ions and act as strong reducing agents. They are soft metals with low melting points and react violently with water to form hydroxides and hydrogen gas.
17. **Correct Answer: B.**
Inert gases have a complete outermost shell (octet, except helium with duplet), which makes them stable and therefore chemically inert.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

18. **Correct Answer: A, D.**
Chromium, manganese and chromium, copper follow the rule of increasing atomic mass.
19. **Correct Answer: A, B, C.**
Correction of atomic weights of some elements is an **achievement**, not a defect.
A, B, and C are genuine defects of Mendeleev's table.
20. **Correct Answer: A, B.**
The law of octaves was the first logical attempt to classify elements based on atomic weights. Periodicity of elements was recognised for the first time.
21. **Correct Answer: A, B, D.**
In a Dobereiner triad, the atomic mass of the middle element is approximately the average of the other two. (Li, Na, K), (Ca, Sr, Ba) and (Cl, Br, I) satisfy this condition.

22. **Correct Answer: C, D.**
Chlorine is the most negative in its group. Sulfur is more negative than Oxygen.
23. **Correct Answer: A, B, D.**
A is correct because this is the definition of electron gain enthalpy. B is correct because Non-metals release energy when gaining electrons. C is incorrect because Down a group, atomic size increases and attraction decreases, so electron gain enthalpy becomes less negative. D is correct because effective nuclear charge increases across a period, making electron gain easier.

ASSERTION AND REASON TYPE

24. **Correct Answer: B.**
Halogens (F, Cl, Br, I, At) are indeed placed in Group 17. The word halogen means salt former, and their salts occur in seawater.
25. **Correct Answer: A.**
Lithium behaves differently from other alkali metals (harder metal, higher melting point, forms Li_2O and Li_3N, etc.). These unusual properties arise due to its small size, high electronegativity, no vacant d-orbitals and strong polarising power.
26. **Correct Answer: C.**
Mendeleev's periodic table made the study of elements easier by grouping elements with similar properties together. Mendeleev arranged elements mainly in order of increasing atomic mass, not atomic number (atomic number arrangement was introduced later by Moseley).
27. **Correct Answer: D.**
Van der Waals radius is also called non-bonded atomic radius and is measured when atoms are not joined by a covalent bond (mainly in inert gases and molecular solids). By definition, van der Waals radius is half the distance between the nuclei of two closest non-bonded atoms in the solid state.
28. **Correct Answer: A.**
As nuclear charge increases across a period, atoms hold electrons more strongly and lose them less easily. This directly explains why metallic character decreases along a period.

MATRIX MATCHING TYPE

29. **Correct Answer: a-Q, b-P, c-S, d-R.**
Helium is the first noble gas and has atomic number 2. Neon is the second noble gas with atomic number 10. Argon has atomic number 18 and belongs to noble gases. Krypton is a noble gas with atomic number 36.
30. **Correct Answer: a-P, b-R, c-S, d-Q.**
Group 1 elements (Li, Na, K, etc.) are called alkali metals. Group 17 elements (F, Cl, Br, I) are known as halogens (salt formers). Group 2 elements (Be, Mg, Ca, etc.) are called alkaline earth metals. Group 18 elements (He, Ne, Ar, etc.) are chemically inert and called noble gases.

COMPREHENSION TYPE

Passage - I

31. **Correct Answer: B.**
Lothar Meyer plotted atomic volume against atomic mass and showed that properties of elements vary periodically.
32. **Correct Answer: B.**
After the peak (alkali metals), the curve slopes downward and then rises again; the ascending part corresponds mainly to alkaline earth metals (Group 2).
33. **Correct Answer: B.**
From alkali metals (peak) to noble gases (trough), the curve falls gradually, and this descending portion is mainly occupied by alkaline earth metals.

Passage - II

34. **Correct Answer: D.**
Mendeleev predicted eka-boron, which was later discovered as scandium (Sc).
35. **Correct Answer: A.**
Eka-aluminium predicted by Mendeleev was later discovered and named gallium (Ga).

36. Correct Answer: C.
Eka-silicon predicted by Mendeleev was later identified as germanium (Ge).

INTEGER TYPE

37. Correct Answer: 32.
The 6th period includes elements from Cs (55) to Rn (86) and contains s-block (2), f-block (14), d-block (10), and p-block (6) elements.
So, total = $2 + 14 + 10 + 6 = 32$.

38. Correct Answer: 2
Magnesium (Mg, Z = 12) has electronic configuration 2, 8, 2.
It loses two valence electrons to attain a stable noble gas electronic configuration, hence its valency is 2.

39. Correct Answer: $10\,\text{Å}$

$$1\,\text{nm} = \frac{10^{-9}}{10^{-10}} = 10\,\text{Å}$$

Previous Year Questions

1. Correct Answer: B.
Among halogens, Cl has more negative electron gain enthalpy than F due to less electron–electron repulsion in larger size → Cl = 349, F = 328. In Group 16, S has more negative value than O → S = 200, O = 141.

2. Correct Answer: C.
Na_2O is an oxide of a metal (sodium) and is basic in nature. SO_2, CO_2, and P_4O_{10} are oxides of non-metals and are acidic oxides.

3. Correct Answer: D.
Group 15 elements have half-filled p-orbitals (p^3), which makes them more stable and gives them higher ionization enthalpy than Group 16 (p^4).

4. Correct Answer: C.
Fluorine has a very small size, so added electron experiences strong repulsion in its compact 2p orbital, making its electron-gain enthalpy less negative than chlorine.
The reason is wrong because electron–electron repulsion is higher in fluorine, not in chlorine.

5. Correct Answer: B.
Order of first ionization enthalpy in 2nd period (approx.):
Li < B < Be < C < O < N < F < Ne
So, second lowest = B, second highest = N.

6. Correct Answer: B.
Total electrons = $2 + 2 + 6 + 2 + 5 = 17$→ element is chlorine (Cl).

7. Correct Answer: B.
Metallic character increases down a group and decreases from left to right across a period.

8. Correct Answer: A.
Across a period (Na → Mg → Al → Si), electronegativity increases due to increasing nuclear charge and decreasing atomic size.

9. Correct Answer: A.
Magnesium has a stable $3s^2$ configuration, while aluminium has $3s^2 3p^1$.
Removing an electron from Al (3p electron) is easier than from Mg (3s electron), so Mg has higher first ionization enthalpy than Al.

10. Correct Answer: C.
All given ions are isoelectronic (18 electrons).
For isoelectronic species, ionic radius decreases with increasing nuclear charge.
Nuclear charge order: S (16) < Cl (17) < K (19) < Ca (20)
So, S^{2-} has the least nuclear charge and hence the largest ionic radius.

3. Atoms and Molecules

Solutions

Worksheet –1

1. **Correct Answer: B**

   Berzelius introduced the modern chemical symbol system using one or two letters (capital–lowercase) derived from element names to standardize notation.

2. **Correct Answer: D**

   The symbol C is derived from the Latin name Carbo, standardized by IUPAC to represent carbon.

3. **Correct Answer: C**

   Mercury's symbol Hg originates from its Latin name Hydrargyrum, meaning "liquid silver."

4. **Correct Answer: A**

   Mendelevium (Md, Z = 101) is named in honor of Dmitri Mendeleev for his periodic table framework.

5. **Correct Answer: D**

   Neptunium was named after the planet Neptune, following the convention of naming transuranic elements after celestial bodies.

6. **Correct Answer: A**

   Germanium (Z = 32) was named Germania by Clemens Winkler in honor of his homeland, Germany; this element notably filled the "ekasilicon" gap predicted by Mendeleev's periodicity.

7. **Correct Answer: C**

   In stoichiometry, the mass of an element numerically equal to its atomic weight but expressed in grams is formally termed the Gram Atomic Weight (GAW), which is functionally equivalent to the mass of one "Gram atom" or one mole of atoms.

8. **Correct Answer: A**
   Relative atomic mass is a dimensionless ratio defined as the average mass of atoms of an element compared to 1/12 of the mass of a Carbon-12 atom; for oxygen, this value is approximately 16.

9. **Correct Answer: C**
   Curium (Z = 96) is a transuranic radioactive element named to honor Marie and Pierre Curie, pioneers in the study of radioactivity and the discoverers of polonium and radium.

10. **Correct Answer: B**
    Po is the chemical symbol for Polonium (Z = 84), a highly radioactive metalloid discovered by Marie Curie and named after her native country, Poland (Polonia).

Worksheet –2

1. **Correct Answer: D**
   GMW = 2 × 35.5 = 71 g/mol

2. **Correct Answer: A**
   Tetravalent ions exhibit a valency of 4, corresponding to a ±4 charge magnitude

3. **Correct Answer: B**
   Molecular mass = 12 + 4(1) = 16 u

4. **Correct Answer: C**

   A molecule is defined as the minimum discrete unit retaining chemical identity and capable of independent existence under normal conditions.

5. **Correct Answer: C**

   A trivalent electropositive ion must carry a +3 charge

6. **Correct Answer: B**

   Atoms form molecules or ions to attain a lower potential-energy electronic configuration, typically corresponding to a noble-gas (octet/duplet) structure.

7. **Correct Answer: A**
An electronegative bivalent ion must carry a −2 charge; carbonate is a polyatomic anion with net charge −2.
8. **Correct Answer: C**
A radical composed of two or more different atoms acting as a single charged entity is defined as a compound (polyatomic) radical.
9. **Correct Answer: A**
A monovalent ion has a unit charge magnitude (±1)
10. **Correct Answer: C**
Atomicity =
number of atoms per molecule;
$P4 \Rightarrow 4\,\text{atoms} \Rightarrow \text{tetra-atomic}$

Worksheet – 3

1. **Correct Answer: A**
Combining capacity is defined as the number of electrons an atom can lose, gain, or share during bond formation, which is its valency.
2. **Correct Answer: C**
$M = 2 \times \text{Vapour Density} \Rightarrow$
$M = 2 \times 35.5 = 71\,\text{g mol}^{-1}$
3. **Correct Answer: A**
Sodium has electronic configuration $1s^2$ $2s^2$ $2p^6$ $3s^1$ so the outermost shell contains one electron.
4. **Correct Answer: A**
44g of CO_2 corresponds to 1 mole of CO_2 which contains 2 moles of oxygen atoms.
5. **Correct Answer: A**
Mercury exhibits variable valency of +1 and +2, with +2 being its higher oxidation state.
6. **Correct Answer: A**
One mole is defined to contain Avogadro's number of elementary entities. $N_A = 6.022 \times 10^{23}\,\text{mol}^{-1}$
7. **Correct Answer: B**
Silver exhibits variable valency (0 and +1), with +1 being its lower stable valency in compounds.
8. **Correct Answer: D**
22 gm of CO_2. Both 23g of NO_2 and 22g of CO_2 contain exactly 0.5 moles of molecules.
9. **Correct Answer: B**
1 mL of ideal gas at STP consists of 2.688×10^{19} molecules. 5 mL of ideal consist of $2.688 \times 10^{19} \times 5$ molecules.
$2.688 \times 10^{19} \times 5 = 1.344 \times 10^{20}$ molecules
10. **Correct Answer: D**
At STP, 1 mole of any gas occupies 22.4 L; two gram-moles correspond to 2 moles of gas.

Worksheet – 4

1. **Correct Answer: C**
A skeletal equation shows the reactants and products of a reaction but is not balanced.
2. **Correct Answer: B**
A single compound, calcium carbonate, breaks down into two simpler substances, calcium oxide and carbon dioxide, which is characteristic of a decomposition reaction.
3. **Correct Answer: B**
Sodium chlorate consists of Na^+ and the chlorate ion (ClO_3^-), giving $NaClO_3$.
4. **Correct Answer: B**
This equation conserves both hydrogen and oxygen atoms and uses molecular forms of H_2 and O_2, which is chemically appropriate.
5. **Correct Answer: B**
X = group 2 element (Ca-like), Y = group 17 element (Cl-like). So compound formed = XY_2.
6. **Correct Answer: B**
A balanced chemical equation directly provides mole ratios between reactants and products. From $CaCO_3 \rightarrow CaO + CO_2$, the mole ratio of $CaCO_3$ to CO_2 is 1 : 1.
7. **Correct Answer: B**
Balancing $CO + O_2 \rightarrow CO_2$ by the partial (inspection) method gives:
$2CO + O_2 \rightarrow 2CO_2$, which conserves both carbon and oxygen atoms.

8. **Correct Answer: B**
A chemical equation represents a chemical reaction using chemical symbols and formulas. It does not inherently show colour, temperature, or only products.
9. **Correct Answer: A**
18 g of water is composed of 2 g of hydrogen and 16 g of oxygen because molecular weight of oxygen and hydrogen is 16 g/mol and 1 g/mol.
Mol. Wt. (H_2O) = Mol. Wt. (O) + 2(Mol. Wt. (H)) = 16 g/mol + 2 g/mol = 18 g/mol
10. **Correct Answer: B**
Adding reaction conditions such as temperature, pressure, and catalyst (e.g., Fe in the Haber process) makes the equation more informative without altering the stoichiometry.

Competitive Corner

SINGLE CORRECT OPTION TYPE

1. **Correct Answer: B**
At STP (0 °C and 1 atm), 1 mole of an ideal gas occupies 22.414 L (often approximated as 22.4 L).
2. **Correct Answer: A**
Cd is the correct chemical symbol for cadmium.
3. **Correct Answer: A**
Electrons in the outermost shell are called valence electrons.
4. **Correct Answer: A**
The Latin name for gold is Aurum
5. **Correct Answer: B**
1 mole of any substance contains Avogadro's number = 6.022×10^{23} particles (commonly rounded to 6.023×10^{23}).
6. **Correct Answer: C**
Molar mass of CH_3COOH = 2C (24) + 4H (4) + 2O (32) = 60 g mol^{-1}
7. **Correct Answer: B**
Atom A (2,8,3) has valency = 3 (loses 3 electrons).
Atom B (2,6) has valency = 2 (gains 2 electrons).
Using criss-cross method → A_2B_3 is correct.
8. **Correct Answer: C**
By the Law of Conservation of Mass:
Mass of reactants = Mass of products = 5 g + 3 g = 8 g
9. **Correct Answer: A**
A bivalent electropositive ion is a metal cation with +2 charge.
10. **Correct Answer: C**
It increases from IA to IVA and then decreases up to VIIA.
11. **Correct Answer: A**
Atomic number 16 corresponds to sulphur (S).
12. **Correct Answer: C**
Except iron, every element is balanced. Fe_3O_4 contains 3 iron atoms, so iron in the left side of the reaction should have a coefficient of 3.
13. **Correct Answer: B**
According to Dalton's atomic theory, compounds are created when atoms link up in fixed, simple whole-number ratios.
14. **Correct Answer: B**
When magnesium burns in air, it forms magnesium oxide, a new substance produced by rearrangement of atoms, showing that a chemical reaction has occurred.
15. **Correct Answer: B**
Fluorine has 7 valence electrons; to attain a stable configuration of 6 electrons, it would lose one electron, forming F^+.
16. **Correct Answer: B**
In option B, zinc being more reactive than copper displaces copper from copper sulphate to form zinc sulphate and free copper, which is the defining feature of a displacement reaction.
17. **Correct Answer: D**
The ions of silver nitrate and sodium chloride exchange partners to form two new compounds, silver chloride and sodium nitrate, showing a double displacement reaction.

18. **Correct Answer: A**
Iron is oxidized because it combines with oxygen to form iron hydroxide/rust.
19. **Correct Answer: A**
A chemical equation shows reactants, products, their chemical formulas, and relative amounts (via coefficients), but it does not convey the time required for the reaction.
20. **Correct Answer: B**
By the hit and trial method, coefficients are adjusted until the number of Al, Cl, and H atoms becomes equal on both sides; this is satisfied only in option (B).

ONE OR MORE THAN ONE CORRECT OPTION TYPE

21. **Correct Answer: A, B, C.**
Mole = amount of substance containing as many entities as atoms in 12 g of C-12. mole contains Avogadro's number (6.022×10^{23}) of entities. Mole is the SI unit of amount of substance.
22. **Correct Answer: A, B, C.**
A symbol is an abbreviation of the full name of an element (e.g., Na for sodium). The method of naming substances is called chemical nomenclature. Representing substances using symbols/formulae is called chemical notation.
23. **Correct Answer: A, B.**
Count atoms of each element on both sides — This is the first step in the hit and trial method to compare reactants and products. Balance the element with the least number first — In the hit and trial method, we usually start balancing the element that appears in the smallest number (or the simplest one) to make adjustment easier.
24. **Correct Answer: A, B, D.**
Cu^{2+} ion is cupric ion. Fe^{2+} ion is ferrous ion. Hg_2^{2+} is mercurous ion
25. **Correct Answer: A, C, D.**
Cl_2, O_2, H_2 are diatomic.
26. **Correct Answer: A, B, C.**
A chemical reaction is indicated by observable changes such as colour change, release of heat or light, and formation of a new substance, whereas a change only in shape or size is a physical change.
27. **Correct Answer: A, B.**
In option A and B, two substances combine to form a single product.
28. **Correct Answer: A, B, D.**
Painting or oiling and galvanizing protect the metal from contact with air and moisture, while alloying (such as making stainless steel) produces a material that is more resistant to rust, so these methods help prevent corrosion.

ASSERTION AND REASON TYPE

29. **Correct Answer: C**
Assertion is true (atomic radius is measured in nm). Reason is false because atoms are not visible to the naked eye; nm is used due to their extremely small size.
30. **Correct Answer: A**
Both are true, and the Reason correctly explains the Assertion: CO_2 means 1 carbon atom and 2 oxygen atoms.
31. **Correct Answer: C**
Assertion is true (some metals show variable valency). Reason is false because variable valency is mainly due to the involvement of both ns and (n–1)d electrons, not because they have 4–7 valence electrons.
32. **Correct Answer: D**
The Assertion is false because a molecule can exist independently. The Reason is true: a molecule is the smallest particle of an element or compound that can exist independently and shows all its properties.

MATRIX MATCHING TYPE

33. **Correct Answer: a-R, b-S, c-P, d-Q.**
These are the Latin names from which the chemical symbols are derived: Pb (Plumbum), Sn (Stannum), Fe (Ferrum), Hg (Hydrargyrum).

34. Correct Answer: a-P, b-Q, c-R, d-S.
Calculated using atomic masses:
H_2O = 2(1) + 16 = 18 u, Cl_2 = 2×35.5 = 71 u, N_2 = 2×14 = 28 u, H_2 = 2×1 = 2 u.

COMPREHENSION TYPE

35. Correct Answer: B
Molar mass of $CuSO_4$ = 64 + 32 + 4×16 = 160 g mol^{-1}
Moles = 1.60 / 160 = 0.01 mol

36. Correct Answer: A
1 gram of hydrogen contains 6 × 10^{23}atoms (i.e., 1 mole of atoms).
Helium has an atomic mass of 4, so 4 grams of He = 1 mole of He atoms.
Therefore, 4 g of He also contains 6 × 10^{23}atoms.

37. Correct Answer: C
Molar mass of $CaCO_3$ = 40 + 12 + 48 = 100 g mol^{-1}
Moles in 50 g = 50/100 = 0.5 mol
Each $CaCO_3$ has 3 O atoms → O atoms = 0.5 × 3 × 6.023 × 10^{23} = 9.035 × 10^{23} atoms

38. Correct Answer: B
In NH_3, nitrogen forms three bonds with three hydrogen atoms, so its valency is 3.

39. Correct Answer: C
In PCl_5, phosphorus forms five bonds with five chlorine atoms, so its valency is 5.

40. Correct Answer: C
In CO_2, carbon forms four bonds (two double bonds with two oxygen atoms), so its valency is 4.

INTEGER TYPE

41. Correct Answer: 2.
Valency of copper in cupric chloride = 2
Copper shows valency of 2 in $CuCl_2$ (cupric chloride).

42. Correct Answer: 14.
The atomic number of silicon is 14.

43. Correct Answer: 6.022×10^{23}.
Avogadro's number is 6.022×10^{23}.

Previous Year Questions

1. Correct Answer: B
Calcium sulfite is $CaSO_3$. Its formula unit mass is $40+32+(16\times3)=120\,u$.
Carbon dioxide has a fixed mass ratio of $C:O=12:32$ which simplifies to 3 : 8.
The Law of Constant Proportions states this remains the same regardless of the preparation method. Ozone is triatomic.
Elemental sulfur is polyatomic. Nitride is an anion, but phosphate is also an anion (negatively charged), not a cation.

2. Correct Answer: B
For CO_2, the fixed mass ratio of carbon to oxygen is:
C : O = 12 : 32 = 3 : 8

3. Correct Answer: C
Zinc has a valency of +2 and sulfate is -2. The formula is $ZnSO_4$.
Phosphorus is tetraatomic (P_4) and chlorine is diatomic (Cl_2).
For Magnesium sulfide, mass ratio is $Mg(24):S(32)=3:4$
Ammonium sulfate has a mass of 132 u. 96 u is only the mass of the sulfate ion.

4. Correct Answer: A
P (Carbon monoxide): 12: 16 = 3: 4
Q (Hydrochloric acid): 1: 35.5 = 1: 35.5
R (Carbon dioxide): 12: 32 = 3: 8
S (Nitrogen dioxide): 14: 32 = 7: 16

5. Correct Answer: A
Statements II and IV are incorrect. Sulfide is divalent negative, and ammonium is a cation.

6. Correct Answer: B
If X forms XCl_3, its valency is 3. Nitride also has a valency of 3. They combine in a 1: 1 ratio. Then formula for nitride of X is XN.

7. Correct Answer: A

$\text{Moles} = \frac{\text{Mass}}{\text{Molar Mass}}$, He = 28/4 = 7; Na = 46/23 = 2; Ca = 60/40 = 1.5

8. Correct Answer: C

$\text{Total atoms} = \text{Moles} \times \text{Atomicity} \times N_A$,
$0.4 \times 3 = 1.2\,\text{mol atoms}$ This is the highest.

9. Correct Answer: B

Gas X is ammonia. Statement II is true. Statement IV is true because 1 mol of water vapour contains $3 \times 6.023 \times 10^{23}$ atoms.

10. Correct Answer: B

Both experiments yield the same percentage/ratio of copper in copper oxide regardless of the source. This explains the Law of Constant Proportions.

4. Matter In Our Surroundings

Solutions

Worksheet – 1

1. **Correct Answer: C) Mass**
   Explanation: Mass tells us how much matter an object contains. It does not depend on shape or size.
2. **Correct Answer: D) Sun**
   Explanation: The Sun gives energy in the form of heat and light without needing a medium. This transfer of energy is called radiation.
3. **Correct Answer: A) Solids**
   Explanation: Solids have a fixed shape and volume, so they are rigid. Liquids and gases can flow and are not rigid.
4. **Correct Answer: D) Very high temperature**
   Explanation: Plasma forms when gases are heated to extremely high temperatures, causing ionization (loss of electrons).
5. **Correct Answer: A) Gases**
   Explanation: Gases expand to fill any container, so they have neither a fixed shape nor a fixed volume.
6. **Correct Answer: C) Solids**
   Explanation: In solids, particles are closely packed and strongly attracted to each other.
7. **Correct Answer: B) Loosely**
   Explanation: Liquid particles are close but not tightly packed like solids, allowing them to move and flow.
8. **Correct Answer: A) a – Solid, b – Liquid, c – Gas**
   Explanation:
   - Iodine → Solid
   - Bromine → Liquid
   - Fluorine → Gas
     (at room temperature)
9. **Correct Answer: A) Solid**
   Explanation: Solids have tightly packed particles with very little space, so they cannot be compressed easily.
10. **Correct Answer: D) Plasma**
    Explanation: In plasma, particles are highly energetic and move freely with minimal collisions, hence it is called a collision-less state.

Worksheet-2

1. **Correct Answer: C) Gas**
   Explanation: Gas particles move freely in all directions and fill the entire container, so they do not settle at the bottom like solids and liquids.
2. **Correct Answer: D) Plasma**
   Explanation: The extremely high temperature in the sun and stars ionizes gases, forming plasma.
3. **Correct Answer: D) Both A and B**
   Explanation: Both temperature and pressure affect the change of matter from one state to another.
4. **Correct Answer: B) Liquid, vapour and solid**
   Explanation:
   - Room temperature → Liquid
   - Above 100 °C → Vapour (gas)
   - Below 0 °C → Solid (ice)
5. **Correct Answer: b) Interconversion of matter**
   Explanation: Interconversion refers to reversible change between states by altering temperature or pressure.
6. **Correct Answer: b) Boiling / Vapourisation**
   Explanation: When a liquid absorbs heat and changes into gas, the process is called vapourisation or boiling.

7. **Correct Answer: b) 100 °C**
   Explanation: Water boils at 100 °C under normal atmospheric pressure.
8. **Correct Answer: b) Cooling**
   Explanation: During evaporation, high-energy particles escape, lowering the average temperature of the remaining liquid.
9. **Correct Answer: b) Decreases**
   Explanation: Lower pressure allows liquid particles to escape more easily, so boiling occurs at a lower temperature.
10. **Correct Answer: c) Deposition**
    Explanation: Deposition is the process where a gas changes directly into a solid without becoming liquid

COMPETETIVE CORNER

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: B)**
   Intermolecular forces of attraction in gases are minimum.
2. **Correct Answer: B) Feeling of cold**
   Explanation: Matter must have mass and occupy space. A feeling does neither.
3. **Correct Answer: B) Liquids can be compressed easily but not gases.**
   Explanation: Gases can be compressed easily, while liquids cannot.
4. **Correct Answer: D) Gases present in the atmosphere**
   Explanation: LPG, oxygen cylinders, and CNG use compressibility. Atmospheric gases are not stored by compression.
5. **Correct Answer: B) Diffusion of gases**
   Explanation: Perfume spreads in air due to diffusion of gas particles.
6. **Correct Answer: B) Iodine**
   Explanation: Iodine is a solid at room temperature, so it has strong intermolecular forces.
7. **Correct Answer: A) Diffusibility**
   Explanation: Gas particles mix uniformly because of diffusion.
8. **Correct Answer: A) Bromine, mercury**
   Explanation: Both are liquids with relatively strong intermolecular attraction.
9. **Correct Answer: B) Close packing of molecules in solids.**
   Explanation: Strong attraction and close packing restrict movement in solids.
10. **Correct Answer: D) Bose–Einstein condensate**
    Explanation: It is formed at extremely low temperatures close to 0 K.
11. **Correct Answer: B) Evaporation**
    Explanation: Evaporation occurs at all temperatures from the surface of the liquid.
12. **Correct Answer: B) Vapour pressure is equal to atmospheric pressure**
    Explanation: At boiling point, vapour pressure equals atmospheric pressure.
13. **Correct Answer: C) Sublimation**
    Explanation: Naphthalene changes directly from solid to gas.
14. **Correct Answer: A) Freezing / Solidification**
    Explanation: Cooling removes heat, causing liquid particles to become fixed.
15. **Correct Answer: C) Condensation point**
    Explanation: At the condensation point, gas loses heat and becomes liquid.
16. **Correct Answer: B) Gas < Liquid < Solid**
    Explanation: Intermolecular forces are weakest in gases, stronger in liquids, and strongest in solids.
    - Gas → particles are far apart → very weak attraction
    - Liquid → particles are closer → moderate attraction
    - Solid → particles tightly packed → strong attraction
17. **Correct Answer: A) 1 and 2 only**
    Explanation: Increase in temperature → More heat → particles move faster → evaporation increases

    Increase in surface area → More exposed particles → faster evaporation

ONE OR MORE THAN ONE OPTION CORRECT TYPE

18. **Correct Answers: A) Melting, B) Boiling / Vapourisation, D) Evaporation**

    Explanation:
    These processes are endothermic, meaning they absorb heat to overcome intermolecular forces.

19. **Correct Answer: A) Evaporation**

    Explanation:
    Evaporation occurs only at the surface of a liquid.
    - Boiling, sublimation, and melting occur throughout the substance.

20. **Correct Answers: A) Frost formation on cold surfaces, B) Snow formation**
    Explanation:
    Deposition is the direct change from gas to solid.
    - Freezing: liquid → solid
    - Condensation: gas → liquid

21. **Correct Answers: A) Are closely packed, C) Vibrate about a fixed position**
    Explanation:
    Solid particles are tightly packed and can only vibrate.
    They cannot move freely and have the least kinetic energy.

22. **Correct Answers: A) Plasma is an ionized gas, B) Plasma occurs naturally in the Sun**
    Explanation:
    Plasma is a high-energy, ionized state of matter found in stars.
    It has no definite shape and forms at very high, not low, temperatures.

23. **Correct Answers: A) Formed at near absolute zero, B) Atoms collapse into a single quantum state, C) Has properties of a superfluid**

    Explanation:
    BEC forms at extremely low temperatures where atoms behave as one quantum entity.
    It does not exist at high temperatures.

ASSERTION AND REASNING TYPE

24. **Correct Answer: A) Both A and R are true, and R is the correct explanation of A**
    Explanation:
    Because particles in a solid are tightly packed and can only vibrate in fixed positions, solids maintain a definite shape and volume.

25. **Correct Answer: D) A is false, R is true**
    Explanation:
    - Boiling occurs only at a fixed temperature (boiling point).
    - The reason statement is also false in reality (boiling is a bulk phenomenon, evaporation is a surface phenomenon).

26. **Correct Answer: A) Both A and R are true, and R is the correct explanation of A**
    Explanation:
    Heating increases kinetic energy and cooling decreases it, enabling particles to overcome or strengthen intermolecular forces, causing state change.

27. **Correct Answer: A) Both A and R are true, and R is the correct explanation of A**
    Explanation:
    At 0 °C, ice melts into water and water freezes into ice, depending on heat gain or loss.

28. **Correct Answer: D) A is false, R is true**
    Explanation:
    Condensation is an exothermic process because heat is released when gas changes into liquid.

MATCH THE FOLLOWING TYPE

29. **Correct Answer: a–Q, b–R, c–P, d–S**

    Explanation:
    - Solid → Q:
      In solids, particles are tightly packed with strong intermolecular forces, giving them a fixed shape and volume.
    - Liquid → R:
      Liquids have particles that are less tightly packed than solids.

They have a fixed volume but take the shape of the container.

- Gas → P:
  Gas particles are very far apart with negligible attraction, so gases have neither fixed shape nor volume.
- Plasma → S:
  Plasma is a highly energized state of matter consisting of free electrons and ions (ionized gas).

30. **Correct Answer: a–S, b–T , c–P, d–Q**
Explanation
    - Melting**:** Solid gains heat and changes into liquid.
    - Freezing: Liquid loses heat and becomes solid.
    - Evaporation: Liquid changes into gas below boiling point from the surface.
    - Sublimation**:** Solid changes directly into gas without becoming liquid.

COMPREHENSHION TYPE

I.

31. **Correct Answer: b) An ionized gas**
Explanation:
Plasma is often called the fourth state of matter. It forms when a gas is given so much energy that its atoms lose electrons, creating charged particles (ions and free electrons)**.** Because of this, plasma can conduct electricity and respond to magnetic fields. Examples include lightning and the Sun.
32. **Correct Answer: C) Near absolute zero**
Explanation:
A Bose-Einstein Condensate (BEC) forms when particles are cooled to temperatures extremely close to absolute zero (0 Kelvin or -273°C)**.** At this point, atoms move very slowly and begin to behave like a single "super-atom" with shared quantum properties.
33. **Correct Answer: B) Stars**
Explanation:
Stars, including the Sun, are made mostly of plasma**.** The extremely high temperatures cause gases to become ionized, turning them into plasma. Other natural examples include lightning and auroras.

II.

34. **Correct Answer: B) At any temperature below boiling point**
Explanation:
Evaporation is **a** surface phenomenon that can happen at all temperatures, not just at the boiling point. Even at room temperature, some faster-moving liquid particles escape into the air as vapor.
35. **Correct Answer: B) Bulk phenomenon**
Explanation:
Boiling occurs throughout the entire liquid**,** not just at the surface. When a liquid reaches its boiling point, vapor bubbles form inside the liquid and rise to the surface, making it a bulk phenomenon**.**
36. **Correct Answer: A) Increase in surface area**
Explanation:
A larger surface area allows more liquid particles to escape into the air at the same time, which increases the rate of evaporation**.**
    - Higher humidity slows evaporation
    - Lower wind speed slows evaporation
    - Lower temperature slows evaporation

INTEGER TYPE:

37. **Correct Answer: 0°C**
Explanation:
Ice melts and turns into liquid water at 0°C under normal atmospheric pressure.
38. **Correct Answer: 100°C**
Explanation:
At 1 atmosphere of pressure, water boils and changes into steam at 100°C.
39. **Correct Answer: 0°C**
Explanation:
Water freezes and turns into ice at the same temperature it melts, which is 0°C.
40. **Correct Answer: 373 K**
Explanation:
To convert Celsius to Kelvin:

K = °C + 273
So, 100°C + 273 = 373 K

Previous year questions

1. Correct Answer: B) II and IV only

Explanation: Ice at 273 K is more effective in cooling than water at 273 K.– Ice absorbs extra heat (latent heat of fusion) while melting, so it cools better than water at the same temperature.

Evaporation and fusion processes involve absorption of heat.
Evaporation absorbs heat

- Fusion (melting) absorbs heat

2. Correct Answer:D) All of these

Explanation:
A substance can exist as:

- Solid below melting point
- Liquid between melting and boiling points
- Gas above boiling point

3. Correct Answer:

D) Hydrogen < Petrol < Wooden block

Explanation:

Rigidity increases as:
Gas < Liquid < Solid

4. Correct Answer: D) Both B and C

Explanation:

Matter is made up of particles
– The salt exists as tiny particles that mix into the water.

Particles of salt get into the spaces between the particles of water

– This is exactly what the experiment demonstrates.

5. Correct Answer: B) Both statement-I and statement-II are not correct

Explanation:

- Strong intermolecular forces mean particles stick together → solid or liquid
- Gases form when thermal energy is greater than intermolecular forces

6. Correct Answer:

Explanation: B) a, b, c

A. Evaporation of liquid
Increases — higher temperature gives particles more energy to escape.

B. Sublimation of solid
Increases — more heat helps solids change directly into gas.

C. Solubility of solute in water (solid in liquid)
Generally increases — most solids dissolve more in warm water.

7. Correct Answer: B) Water

Explanation:

Water can exist as:

- Solid → Ice
- Liquid → Water
- Gas → Water vapor/steam

8. Correct Answer: B) Carbon dioxide

Explanation:

In the lungs:

- Oxygen diffuses from lungs → blood
- Carbon dioxide diffuses from blood → lungs (to be exhaled)

9. Correct Answer: C) R

Explanation:

For a substance to be:

- Liquid at −140°C → Melting point must be below −140°C
- Gas at −100°C → Boiling point must be below −100°C

10. Correct Answer: B) −100°C to −95°C

A substance exists as a liquid between its melting and boiling points

So liquid range is: −110°C to −85°C

5. Is Matter Around Us Pure?

Solutions

Worksheet – 1

1. **Correct Answer: A.**
   A pure substance (element or compound) consists of only one type of particle or chemical entity, resulting in a single-phase system with invariant chemical and physical properties throughout its bulk.
2. **Correct Answer: B**
   Air is a multi-component system where the atmospheric gases retain their individual chemical identities and exist in variable proportions.
3. **Correct Answer: B**
   Heterogeneous mixtures exhibit spatial variations in concentration and physical state.
4. **Correct Answer: B**
   Brass is a substitutional solid solution where solute atoms (Zn) occupy lattice sites of the solvent (Cu) within a single FCC or BCC metallic phase, maintaining macro-scale uniformity.
5. **Correct Answer: C**
   Gunpowder is a mechanical aggregate of potassium nitrate, charcoal, and sulfur that can be separated via physical solvent extraction.
6. **Correct Answer: C**
   Milk is a colloidal dispersion (emulsion) consisting of liquid fat globules dispersed in an aqueous phase; although it appears uniform to the naked eye, it is multi-phasic at the microscopic scale.
7. **Correct Answer: C**
   Compounds are formed when atoms achieve a lower energy state through chemical bonding in a fixed stoichiometric ratio, dictated by the Law of Definite Proportions.
8. **Correct Answer: C**
   Oil and water are immiscible due to the high interfacial tension and the disparity in polarity, leading to a system with two distinct liquid phases.
9. **Correct Answer: B**
   Charcoal (C) and Salt (NaCl) are both solids that do not form a solid solution; they remain as a physical aggregate of two distinct solid phases.
10. **Correct Answer: B**
    Mercury (Hg) has a low melting point (234.3 K).

Worksheet – 2

1. **Correct Answer: B**
   The Tyndall effect occurs when the dispersed phase particles in a colloid have dimensions comparable to the wavelength of incident visible light, causing elastic scattering (Rayleigh or Mie scattering depending on particle size) that renders the beam path visible.
2. **Correct Answer: B**
   By scientific classification, any mixture with a dispersed phase in the 1 nm to 1000 nm range is defined as a colloid.
3. **Correct Answer: B**
   Sodium chloride in water forms a true solution. True solution does not show Tyndall effect.
4. **Correct Answer: A.**

$$\%(w/v) = \frac{\text{Mass of Solute (g)}}{\text{Volume of Solution (mL)}} \times 100$$

$$\Rightarrow 10\% = \frac{10\ \text{g}}{100\ \text{mL}} \times 100$$

5. **Correct Answer: B**
   Strength (S) is a quantitative measure of concentration representing the mass of solute (m) present in a unit volume (V) of solution, typically expressed in g/L.

6. **Correct Answer: C**
Ethanol and water are completely miscible liquids across all proportions due to the formation of extensive intermolecular hydrogen bonding, resulting in a liquid-in-liquid true solution (homogeneous mixture).
7. **Correct Answer: B**
Benzene and toluene are non-polar organic liquids that exhibit nearly ideal mixing behavior because their structural similarities lead to comparable London dispersion forces.
8. **Correct Answer: A**
Occlusion refers to the adsorption and subsequent absorption of gas molecules (hydrogen) into the interstitial sites of a solid metallic lattice (palladium), forming a solid solution where gas is the solute.
9. **Correct Answer: D**
Many cough syrups are formulated as pharmaceutical suspensions where finely divided solid drug particles exceed the colloidal size limit (> 1000 nm) and are dispersed in a liquid vehicle, eventually settling under gravity if not stabilized by flocculating agents.
10. **Correct Answer: B**
Camphor acts as a solid dispersed phase that sublimes into the atmosphere (dispersion medium), creating a heterogeneous system categorized as a solid aerosol.

Worksheet – 3

1. **Correct Answer: D**
This process relies on the gravitational force acting on suspended particles whose density exceeds that of the continuous phase, leading to terminal velocity settling.
2. **Correct Answer: C**
Decantation is the macroscopic mechanical separation of a supernatant liquid from a solid precipitate or an immiscible liquid phase by utilizing the difference in spatial distribution after sedimentation.
3. **Correct Answer: C**
This phase transition exploits the disparity in vapor pressures between the non-volatile solute (NaCl) and the volatile solvent (H_2O), where thermal energy overcomes the enthalpy of vaporization of the solvent.
4. **Correct Answer: A**
Distillation integrates vaporization followed by condensation, allowing for the recovery of both the volatile distillate and the non-volatile residue.
5. **Correct Answer: A**
In the context of phase equilibria, the term "sublimate" refers specifically to the solid deposit formed via desublimation (deposition) when a gas bypasses the liquid phase upon cooling below the triple point pressure.
6. **Correct Answer: A**
This separation is possible because Iodine possesses a high vapor pressure at temperatures below its melting point, allowing it to transition directly to the gas phase, while sand reains solid due to its high lattice energy.
7. **Correct Answer: A**
Centrifugation accelerates the sedimentation process by substituting the gravitational field with a centrifugal field, significantly increasing the sedimentation rate for small or low-density-difference particles
8. **Correct Answer: C**
Biological fluids like blood are colloidal suspensions; centrifugation uses differential sedimentation velocities to separate cellular components (pellet) from plasma (supernatant)
9. **Correct Answer: B**
Handpicking is a macro-scale separation technique based on visual and morphological descriptors (size, shape, color) of heterogeneous mixtures where the components are large enough for manual manipulation

10. **Correct Answer: D**
Sieving utilizes a physical barrier (mesh) with defined apertures to separate a polydisperse particulate mixture into size fractions based on a geometric "pass/fail" criterion relative to the mesh diameter.

Worksheet – 4

1. **Correct Answer: C**
Dissolution of glucose is a physical change involving the disruption of intermolecular hydrogen bonds and van der Waals forces to form a homogeneous mixture, without altering the molecular identity of the solute.
2. **Correct Answer: A**
This phenomenon is an enzymatic browning reaction where polyphenol oxidase (PPO) catalyzes the oxidation of phenolic compounds into ortho-quinones, which subsequently polymerize to form brown melanin pigments.
3. **Correct Answer: B**
Cooking involves complex biochemical transformations, including the Maillard reaction (non-enzymatic browning) and protein denaturation, resulting in the formation of new covalent bonds and flavor compounds.
4. **Correct Answer: A**
Chemical changes are generally characterized by high activation energy barriers and significant changes in Gibbs free energy, making the reformation of original reactants energetically unfavorable under standard conditions.
5. **Correct Answer: B.**
A chemical change is defined by the rearrangement of atomic connectivity and the breaking/forming of chemical bonds, leading to a distinct electronic structure and macroscopic properties.
6. **Correct Answer: A**
Dew formation is a phase transition (condensation) where latent heat of vaporization is released as water vapor transitions to the liquid state due to a temperature drop below the dew point.
7. **Correct Answer: A**
Boiling is a physical phase transition occurring when the vapor pressure of the liquid equals the external atmospheric pressure, involving only a change in state and not molecular composition.
8. **Correct Answer: B**
Evaporation is a surface phenomenon where molecules with sufficient kinetic energy overcome intermolecular attractive forces to escape into the gas phase.
9. **Correct Answer: C**
Blood clotting involves a complex cascade of chemical reactions, specifically the conversion of soluble fibrinogen into insoluble fibrin polymers via the enzyme thrombin.
10. **Correct Answer: C**
The enthalpy change of a reaction determines whether energy is evolved or absorbed during the breaking and forming of chemical bonds.

COMPETITIVE CORNER

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: C**

$$\text{Volume\%} = \frac{V_{\text{solute}}}{V_{\text{solute}} + V_{\text{solvent}}} \times 100 \Rightarrow$$

$$\text{Volume\%} = \frac{25\ \text{mL}}{25\ \text{mL} + 250\ \text{mL}} \times 100 = \frac{25}{275} \times 100 \approx 9.09\%$$

2. **Correct Answer: C**
A solution represents a single-phase system where the components are distributed at the molecular or ionic level, satisfying the criteria for thermodynamic homogeneity.
3. **Correct Answer: A**
The nomenclature of solutions is determined by the number of distinct chemical species present; a two-component system specifically defines a binary mixture.

4. **Correct Answer: A**
By IUPAC convention in liquid-phase mixtures, the component present in the sub-stoichiometric or lesser molar/volumetric proportion is designated as the solute.
5. **Correct Answer: B**
A true solution is characterized by a Gibbs phase rule parameter of P = 1, as the solute particles are small enough (< 1 nm) to remain uniformly dispersed without a discernible phase boundary.
6. **Correct Answer: C**
While options A, B, and D are characteristics of many solutions, "Homogeneity" is the fundamental defining property, ensuring uniform composition and properties throughout the entire volume of the phase.
7. **Correct Answer: A.**
In gaseous solutions like air, the component with the highest mole fraction (Nitrogen ≈ 78%) is designated as the solvent, while others are solutes.
8. **Correct Answer: B**
Physical changes involve modifications in intermolecular forces or state without altering the primary chemical bonds or molecular identity, allowing for reversibility via simple thermodynamic shifts.
9. **Correct Answer: C**
Physical changes involve enthalpy changes; endothermic processes (like melting) absorb energy, while exothermic processes (like freezing) evolve energy to the surroundings.
10. **Correct Answer: A**
The hallmark of a physical change is the preservation of the chemical identity and stoichiometry of the substance, whereas chemical changes involve the breaking and forming of covalent or ionic bonds.
11. **Correct Answer: C**
Suspensions are thermodynamically unstable, heterogeneous systems where the dispersed phase particles (typically > 1000 nm) undergo sedimentation due to gravitational force exceeding the kinetic energy of Brownian motion.
12. **Correct Answer: B**
Threshing involves the mechanical rupture of the abscission zone between the grain and the rachis/stalk through impact or rubbing forces, effectively decoupling the edible kernel from the structural plant matter.
13. **Correct Answer: B**
Winnowing utilizes the differential terminal velocities of components in an airflow; the lower density/mass husk experiences a higher drag-to-weight ratio compared to the grain, causing spatial segregation.
14. **Correct Answer: B**
The filtrate is the fluid phase that permeates the porous medium (filter) under a pressure gradient, while the retentate or residue is the solid phase excluded by the pore size cutoff.
15. **Correct Answer: C**
Nitrogen has the lowest boiling point (–196 °C), so it boils off and is collected first.
16. **Correct Answer: B**
Components move at different rates depending on how soluble they are in the solvent (mobile phase) and how strongly they are attracted to the paper (stationary phase).
17. **Correct Answer: B**
Kerosene and water are immiscible and form two layers. Since water is denser, it forms the lower layer and can be separated by opening the stopcock of a separating funnel.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

18. **Correct Answer: C**
Solutions are a homogeneous mixture.
19. **Correct Answer: B, C, D**
Elements do not change structure upon physical division. Elements consist of only one type of atom. Elements cannot be broken down by chemical means

20. **Correct Answer: A, B, C, D**
All four listed elements exist as gases at standard temperature and pressure.
21. **Correct Answer: A, B, C, D**
All four are physical properties characteristic of most metals.
22. **Correct Answer: A, C, D**
Supersaturated solutions are unstable or metastable. Hence the option B is only incorrect statement.

ASSERTION AND REASON TYPE

23. **Correct Answer: A**
The Reason explains the various ways the "100 parts" mentioned in the Assertion can be measured (mass or volume).
24. **Correct Answer: D**
In an open system, it reacts with and gains mass from oxygen in the air.
25. **Correct Answer: C**
Clear water can be separated from muddy water via sedimentation/decantation.
26. **Correct Answer: A**
Light path is not visible in a true solution. Solution particles are too small to scatter light.
27. **Correct Answer: B**
Graphite is an exception among non-metals because of good conductivity of electricity.

MATRIX MATCHING TYPE

28. **Correct Answer: a-Q, b-P, c-S, d-R**
Carbon is a typical non-metal with properties like poor conductivity. Potassium is an alkali metal, highly reactive and metallic. Germanium shows properties of both metals and non-metals, so it is a metalloid. Bromine is the only non-metal that exists as a liquid at room temperature.
29. **Correct Answer: a-Q, b-R, c-S, d-P**
A concentrated sugar solution is called syrup. Salt dissolved in water forms brine. Ammonia gas dissolved in water is known as aqueous ammonia.Carbon dioxide dissolved in water forms seltzer.

COMPREHENSHION TYPE

Passage - I

30. **Correct Answer: B**
In a solution where a solid is dissolved in a liquid, the component that retains its physical state (the liquid) is traditionally defined as the solvent.
31. **Correct Answer: C**
Because the particles are dissolved at the molecular or ionic level, they do not settle under the influence of centrifugal force.
32. **Correct Answer: C**
Options A, B, and D are all defining characteristics of true solutions, whereas C is the opposite of what a solution is.

Passage – 2

33. **Correct Answer: C**
Since both the solute (15 g) and the total solution (100 g) are given in units of mass (grams), the concentration is a mass-by-mass percentage.
34. **Correct Answer: C**
Weight (grams) by volume (milliliters) is the correct nomenclature for this ratio.
35. **Correct Answer: B**
A 20% (w/w) solution means there are 20 units of mass of solute for every 100 units of mass of the total solution.

INTEGER TYPE

36. **Correct Answer: 10%.**
$$\text{Volume\%(v/v)} = \left(\frac{\text{Volume of solute}}{\text{Total volume of solution}}\right) \times 100$$
$$= \left(\frac{25\ \text{mL}}{250\ \text{mL}}\right) \times 100 = 10\%$$
37. **Correct Answer: 15%.**
$$\text{Volume/Weight\%(v/w)} = \left(\frac{15\ \text{mL}}{100\ \text{g}}\right) \times 100 = 15\%$$
38. **Correct Answer: 8:1.**

H_2O water contains oxygen and hydrogen in the ratio of 8:1 by weight.

Previous Year Questions

1. **Correct Answer: C**
   Compounds: (i) Camphor, (ii) Dry ice (CO_2), and (vi) Ammonium chloride (NH_4Cl) have fixed chemical formulas. Mixtures: (iii) Soil, (iv) Aerated drink, (v) Brass (alloy), and (vii) Tincture of iodine (solution) are physical blends.
2. **Correct Answer: A**
   II and III are correct. I is incorrect (milk is an emulsion). IV is incorrect (milky glass is a solid sol—solid in solid).
3. **Correct Answer: D**
   A = $\frac{8}{8+160}\times 100 \approx 4.76\%$, B = $\frac{50}{50+250}\times 100 \approx 16.67\%$, C = $\frac{30}{30+190}\times 100 \approx 13.64\%$, D = $\frac{3}{3+90}\times 100 \approx 3.22\%$
4. **Correct Answer: D**
   Options A, B and C are true characteristics of a compound.
5. **Correct Answer: C**
   A = $\frac{10}{50}\times 100 = 20\%$, B = $\frac{25}{100}\times 100 = 25\%$, C = $\frac{30}{50}\times 100 = 60\%$, D = $\frac{60}{200}\times 100 = 30\%$
6. **Correct Answer: C**
   Colloids: Shaving cream (foam), Clouds (aerosol), Smoke (aerosol), Face cream (emulsion). These are (i), (ii), (iv), and (vi). Suspensions: Chalk powder and water, muddy water. These are (iii) and (v).
7. **Correct Answer: B**
   The Venn diagram shows Y as a colloid where the dispersed phase is liquid and the dispersion medium is solid. This type of colloid is called a gel.
8. **Correct Answer: C**
   This method is used to separate two miscible liquids with a significant difference in their boiling points (typically greater than 25 K).
9. **Correct Answer: C**
   (I) Blood and (III) Milk are colloids; they will scatter light. (II) Brine and (IV) Soda water are true solutions; they will not scatter light.
10. **Correct Answer: D**
    The examples given are Smoke and Shoe polish. This is incorrect. Smoke is a solid dispersed in gas.
11. **Correct Answer: C**
    Statement 1 is true and statement 2 is false

6. Study of Gaseous Laws

Solutions

Worksheet –1

1. **Correct Answer: D) gases**
   Explanation**:** In solids and liquids, particles are packed closely together with very little free space. In **gases**, particles are far apart. This large amount of "empty space" allows them to be easily squeezed (compressed) into a smaller volume when pressure is applied.
2. **Correct Answer: C) Both 1 & 2**
   Explanation**:** Boyle's Law states that for a fixed amount of gas at a constant temperature, the volume (V) is inversely proportional to the pressure (P).
   - Option A ($V \alpha \frac{1}{P}$): This is the direct mathematical statement of inverse proportionality.
   - Option B (PV = K): If you multiply both sides of the proportionality by P, you get PV equals a constant (K).
3. **Correct Answer: B) Hyperbola**
   Explanation: Since P and V are inversely proportional, as one increases, the other decreases. When you plot this relationship on a graph, it forms a smooth, downward-sloping curve technically known as a rectangular hyperbola
4. **Correct Answer: D) Isotherms**
   Explanation: * Iso means "same" and therm refers to "temperature."
   - Isobars are for constant pressure, and isochores are for constant volume.
5. **Correct Answer: B**
   Explanation: Graph B shows the correct hyperbolic curves. Note that as temperature increases ($T_3 > T_2 > T_1$), the curve shifts further away from the origin because the gas occupies more volume at higher temperatures for the same pressure.
6. **Correct Answer: D) Isobars**
   Explanation: The term comes from the Greek word baros, meaning weight or pressure. Just like "isotherms" are for constant temperature, isobars are lines on a graph representing states at the same pressure.
7. **Correct Answer: A)** $V \alpha T$
   Explanation: Charles's Law states that for a fixed mass of gas at constant pressure, the volume (V) is directly proportional to its absolute temperature (T) measured in Kelvin. If you heat a gas up, it expands; if you cool it down, it contracts.
8. **Correct Answer: C) 450 K**
   Explanation: This uses Gay-Lussac's Law ($\frac{P_1}{T_1} = \frac{P_2}{T_2}$) because the volume (10 L) is constant.
   1. Convert Temperature to Kelvin**:** $T_1 = 27°C + 273 = 300\ K$.
   2. Set up the equation**:** $\frac{2.0\,atm}{300K} = \frac{3.0\,atm}{T_2}$.
   3. Solve for T_2**:** $T_2 = \frac{3.0X300}{2.0} = \frac{900}{2} = 450K$
9. **Correct Answer: C) Isochores**
   Explanation: The term isochore refers to a process or graph where the volume remains constant.
   - *Isotopes* relate to atoms with different neutron counts.
   - *Isotheres* (likely a typo for isotheres/isotherms) relate to temperature.

10. Correct Answer: C) Equal volumes of gas contain equal number of moles under similar conditions of temperature and pressure.

Explanation: Avogadro's Law provides the link between the amount of gas (moles, n) and its volume (V). It states that at constant temperature and pressure, the volume is directly proportional to the number of moles ($V \alpha n$). Essentially, if you have two different gases in the same size container at the same temperature and pressure, they contain the exact same number of molecules/moles.

Worksheet-2

1. Correct Answer: B)

Explanation: For one mole (one-gram molecular weight) of an ideal gas, the ideal gas equation is:

PV = nRT

If n = 1, then $\frac{PV}{T} = R$

The Universal Gas Constant (R) in heat units (Calories) is approximately 2 Cal/mol .K.

2. Correct Answer: B)

Explanation:

Convert Units to SI/Standard Gas Units

- Temperature (T): Gas laws require temperature in Kelvin (K).

T = 273°C + 273 = 546 K

Number of moles (n):

$$n = \frac{Mass}{molar\ mass} = \frac{6g}{2g/mol} = 3\,moles$$

Apply the Ideal Gas Equation

The formula is:

PV = nRT

Where R (the universal gas constant) is 0.0821 L. atm.K^{-1}.$mol^{-1.}$

Calculate the Volume (V)

Rearrange the formula to solve for V:

$$V = \frac{nRT}{P}$$

Substitute the values:

$$V = \frac{3X0.0821X546}{1.5} \; ; \; V = \frac{134.4756}{1.5} \; ; \; V \approx 89.65L$$

3. Correct Answer: A)

Explanation:

The term $\frac{PV}{RT}$ represents the Compressibility Factor (Z). For an ideal gas, PV = RT for one mole, which means: $\frac{PV}{RT} = 1$

This value remains constant (1) regardless of changes in temperature or pressure. Therefore, the graph is a horizontal straight line.

4. Correct Answer: D)

Explanation:

By definition, a gas that follows the gas laws (Boyle's, Charles's, and Avogadro's) perfectly under all conditions of temperature and pressure is a theoretical construct.

5. Correct Answer: D)

Explanation:

Real gases deviate from ideal behavior because of intermolecular forces and the volume of the molecules themselves. They behave most like ideal gases when:

High Temperature**:** Kinetic energy is high enough to overcome intermolecular attractions.

Low Pressure**:** Molecules are far apart, making their individual volume negligible compared to the total container volume.

6. Correct Answer: A) NO

Explanation: Step 1: Calculate moles (n)

$$n = \frac{Given\,volume}{Molar\,volume} = \frac{5.6L}{22.4L/mol} = 0.25\,moles$$

Step 2: Find Molar Mass (M)

$$M = \frac{Mass}{moles} = \frac{7.5g}{0.25\,mol} = 30g/mol$$

Step 3: Match with options

A) NO: 14 + 16 = 30

7. **Correct Answer: C) 149.4 lit**
   Explanation:
   $V = \frac{nRT}{P} = \frac{5X0.0821X546}{1.5} \approx$ 149.4 L
8. **Correct Answer: B) 2: 1**
   Explanation:
   Relationship: Mass (m) = n X Molar Mass (M). Since n is constant:
   $\frac{m_1}{m_2} = \frac{M_1}{M_2}$
   Molar Masses:
   - PH_3 = 31 + 3 = 34
   - NH_3 = 14 + 3 = 17

   Ratio: 34: 17 = 2: 1
9. **Correct Answer: D) All of these**
   Explanation:
   The value and units of R depend on the units used for P, V, and Energy.
   A. erg $K^{-1} \cdot mole^{-1}$: Correct (CGS units for energy).
   B. lit.atm $K^{-1} \cdot mole^{-1}$: Correct (Commonly 0.0821).
   C. Cal $K^{-1}.mole^{-1}$: Correct (Approximately 1.987).
10. **Correct Answer: D) 273°C and 1 atm**
    Explanation:
    Density is directly proportional to Pressure (P) and inversely proportional to Temperature (T). To get the *least* density, we need the lowest pressure and the highest temperature

COMPETETIVE CORNER

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: C) They take shape of the container.**
   Explanation: Gases have no fixed shape or volume; they expand to fill whatever space they occupy.
2. **Correct Answer: B) Kelvin**
   Explanation: Kelvin is the base S.I. unit for thermodynamic temperature. Pascal is for pressure; Fahrenheit/Centigrade are common but not S.I.
3. **Correct Answer: A) Absolute Zero**
   Explanation: Absolute Zero (0 K or -273.15°C) is the theoretical lowest temperature where all molecular motion stops.
4. **Correct Answer: D) a, b, c, d**
   Explanation: To describe the state of a gas, we must measure its mass, volume, pressure, and temperature.
5. **Correct Answer: A) I, II, III, IV**
   Explanation: All statements are correct. Matter has three primary states, gases are highly compressible and lack fixed shape/volume, and "vapour" refers to the gaseous phase of substances that are usually solid or liquid at room temperature.
6. **Correct Answer: B) Pressure**
   Explanation: Torr is a unit of pressure based on an absolute scale (1 atm = 760 Torr).
7. **Correct Answer: C) Measurable properties**
   Explanation: Gas laws (like Boyle's or Charles's Law) describe the mathematical relationships between pressure, volume, temperature, and moles.
8. **Correct Answer: C) P/3**
   Explanation: According to Boyle's Law ($P_1V_1 = P_2V_2$), pressure and volume are inversely proportional. If volume triples, pressure must drop to one-third.
9. **Correct Answer: B**
   Explanation: This question refers to **Charles's Law**, which states that for a fixed mass of gas at constant pressure, the volume (V) is directly proportional to its absolute temperature (T).
   - Mathematically: $V \alpha T \text{ or } V = kT$
   - This relationship follows the linear equation $y = mx$.
   - Therefore, a plot of V against T is a straight line passing through the origin.
10. **Correct Answer: A**
    Explanation: Again, we look at Charles's Law ($V/T = k$, where k is a constant).
    - Since the ratio V/T is always equal to a constant value at constant pressure, it does **not** change regardless of how much you increase or decrease the temperature (T).

- A graph of a constant value against a variable results in a horizontal straight line parallel to the x-axis.

11. Correct Answer: A) 0.5

Explanation:

To find the number of moles (n), we use the Ideal Gas Law:

PV = nRT

Calculation:

1. Rearrange the formula to solve for n: $n = \frac{PV}{RT}$
2. Plug in the values: $n = \frac{1X22.4}{0.0821X546}$
3. $n \approx \frac{22.4}{44.8}$
4. $n = 0.5$ moles.

12. Correct Answer: C) 44.8 lit

Explanation: Standard Molar Volume is 22.4 L at $0^{\circ}C$ (273 K). Since the temperature is $273^{\circ}C$ (which is 546 K), the temperature has doubled. According to Charles's Law ($V \alpha T$), the volume must also double.

Calculation: $22.4 \text{ X } 2 = 44.8$ L.

13. Correct Answer: A) Low temperature and high-pressure

Explanation: Real gases behave most like ideal gases at high temperature and low pressure. Therefore, they deviate most when the molecules are forced together and slowed down.

14. Correct Answer: C) $273^{\circ}C$

Explanation: 16g of O_2 is only 0.5 moles (since the molar mass of O_2 is $32 g/mol$). One mole occupies 22.4 L at $0^{\circ}C$ (273 K). For 0.5 moles to occupy that same volume, the temperature must double to expand the gas.

Calculation: 273 K X 2 = 546 K or $273^{\circ}C$.

15. Correct Answer: C) 400 K

Explanation: Using the formula d = $\frac{PM}{RT}$ we see that density is inversely proportional to temperature ($d \alpha \frac{1}{T}$) when pressure is constant.

Calculation: $d_1 T_1 = d_2 T_2$ d X 300K = 0.75d XT_2 T_2 = $\frac{300}{0.75}$ = 400 K

16. Correct Answer: C) 1.5 atm

Explanation: Moles of O_2 = $\frac{8}{32}$ = 0.25

- Moles of N_2 = $\frac{14}{28}$ = 0.50
- Total moles (n) = 0.75
- At STP ($0^{\circ}C$, 1 atm), 1 mole occupies 22.4L. Here, V = 11.2L (half the volume), so pressure would double to 2 atm for 1 mole. Since we have 0.75 moles: P = $\frac{nRT}{V}$ = $\frac{0.75X0.0821X273}{11.2}$ = 1.5 atm.

17. Correct Answer: C) 8.314 K^{-1}. $mole^{-1}$

Explanation: The SI unit for energy is Joules. R = 8.314 $J.K^{-1}mol^{-1}$. Note: Option C is missing the "J", but it is the only one with the correct numerical value for standard SI units.

18. Correct Answer: B) dynes.cm^{-2}; cm^3

Explanation: Ergs are CGS units (centimeters, grams, seconds). Therefore, Pressure must be in dynes.cm^{-2} and Volume in cm^3.

19. Correct Answer: B) Only I, II, III (IV is wrong because the exponent for ergs should be 10^7)

Explanation:

I: 0.0821 $lit.atm.K^{-1}.mol^{-1}$ (Correct)

- II: 62.4 $lit.mm.\ K^{-1}mol^{-1}$
- III: 82.1 $ml.atm.K^{-1}mol^{-1}$ (Correct: 0.0821 \times 1000)

20. Correct Answer: A) dyne.cm

Explanation: Work (Energy) = Force X Distance. In CGS, the unit of Force is the dyne and Distance is the cm.

21. Correct Answer: C) Collision of gas molecules with the walls of the container

Explanation:

Gas molecules are in constant random motion. When they hit the walls of the container, they exert a force. This force per unit area is called pressure.

22. Correct Answer: C) Increases

Explanation:

When volume decreases, the same number of molecules are forced into a smaller space. This makes them hit the container walls more frequently, increasing pressure.

23. Correct Answer: C) Absolute temperature

Explanation:

When temperature increases, gas molecules move **faster** and need **more space**, so the volume increases.

24. Correct Answer: C) 273

Explanation:

Formula:

$$K = °C + 273$$
$$K = 37 + 273 = 310\,K$$

25. Correct Answer: B) Water vapour has lower molecular mass

Explanation:

Water vapour molecules (H_2O) are **lighter** than nitrogen (N_2) and oxygen (O_2), which make up most of dry air. So, when water vapour increases, the air becomes **less dense and lighter**.

26. Correct Answer: C) A straight line with a slope of -1

Explanation:

Boyle's Law: $P \propto \frac{1}{V} \Rightarrow PV = \text{constant}$

Take logarithm on both sides: $P \propto \frac{1}{V} \Rightarrow P = \frac{k}{V}$

Taking log: $\log P = \log k - \log V$

Compare with straight line equation $y = mx + c$

27. Correct Answer: B) 2 : 3

Explanation: $2H_2+O_2 \rightarrow 2H_2O(g)$

Initial moles

- $H_2 = 2\text{moles}$
- $O_2 = 1\text{mole}$

Total initial moles = **3 moles**

After reaction

From the equation:

- $2\text{moles}\ H_2 + 1\text{mole}\ O_2 \rightarrow 2\text{moles}\ H_2O(g)$

All reactants are perfectly used (no excess)

Final moles = 2 moles of water vapor

Step 4: Use gas-volume relation

At same temperature and pressure:

Avogadro's Law

Volume ∝ Number of moles

Step 5: Compare volumes

Initial volume ∝ 3 moles

Final volume ∝ 2 moles

Final ratio (Final : Initial)

$$\frac{2}{3}$$

ONE OR MORE THAN ONE CORRECT OPTION TYPE

28. Correct Answer: All options (A, B, C, D) are correct.

Explanation: A) 0.0821 L.atm.K-1.mole-1: Correct; this is the standard value using liters and atmospheres.

B) 1.99 $\text{cal.K}^{-1}\text{mole}^{-1}$: Correct; often rounded to $2\ \text{cal}$ for simplicity.

C) 8.314 $\text{Kpa.dm}^3\text{.K}^{-1}\text{.mole}^{-1}$: Correct; since $1\ \text{dm}^3 = 1\ \text{L}$ and $1\ \text{Kpa.L}$ is equivalent to Joules.

D) 8.314 $\text{J.K}^{-1}\text{mole}^{-1}$: Correct; this is the standard SI unit value.

29. Correct Answer: A) Volume increases if ideal nature is considered and 'P' & 'n' are kept constant.

Explanation: According to Charles's Law ($V \alpha T$ at constant P and n), if temperature increases, volume must increase.

30. Correct Answer: A) Their pressures are same if 'Y' is of double the size of 'X'.

Explanation: Using $P = \frac{nRT}{V}$:

- For X: $P_X = \frac{nRT}{V_X}$
- For Y: $P_Y = \frac{nRT}{V_Y}$
- If $V_Y = 2V_X$ (double the size), then $P_Y = \frac{2nRT}{2V_X} = \frac{nRT}{V_X} = P_X$.

31. Correct Answer: A and B are both mathematically correct representations.

Explanation: A)PV= $\frac{N}{N_A}RT$: Correct, as $n = \frac{N}{N_A}$ (number of molecules / Avogadro's number).

B) $PV = NK_BT$: Correct, where K_B is the Boltzmann constant (R/N_A).

32. Correct Answer: B) Velocity

Explanation: The equation involves Pressure (P), Volume (V), moles (n), and Temperature (T). Density (\rho) can be derived ($P = \frac{\rho RT}{M}$), but Velocity is not a variable in this macroscopic equation.

33. Correct Answer: All (A, B, C, and D) are units of pressure.

Explanation: Pa (Pascal) is the SI unit. **$N\ m^{-2}$** is the definition of a Pascal. **Bar** and **atm** are common metric and standard units.

34. Correct Answer: B) $\left\{\frac{dV}{dT}\right\}_P = K$

Explanation: charles's Law states $V \alpha T$, or $\frac{V}{T}$ = K (constant).

- Differentiating $\frac{V}{T}$ = K with respect to T gives $\frac{1}{T}\left\{\frac{dV}{dT}\right\} - \frac{V}{T^2} = 0$, which rearranges to $\left\{\frac{dV}{dT}\right\}_P = K$ = K.

35. Correct Answer: A, B, and D.

Explanation:

A: True; gas volume is highly sensitive to P and T.

B: True; solids and liquids are relatively incompressible compared to gases.

D: True; diffusion is the movement of gas particles from high to low concentration.

36. Correct Answer: A, B, and C.

Reasoning: Litre, Millilitre, and dm^3 are all units of volume. Gram is a unit of mass.

ASSERTION AND REASON TYPE

37. Correct Answer: Both (A) and (R) are true, and (R) is the correct explanation of (A).

Explanation:

Assertion (A): Correct. This is the definition of **Boyle's Law**.

Reason (R): Correct. When volume decreases, the gas molecules are crowded into a smaller space, leading to more frequent collisions with the container walls, which manifests as increased pressure.

38. Correct Answer: Both (A) and (R) are true, and (R) is the correct explanation of (A).

Explanation:

Assertion (A): Correct. Charles' Law is expressed as $V \alpha T$ or $\frac{V}{T}$ = k. ($\frac{V_1}{T_1} = \frac{V_2}{T_2}$).

Reason (R): Correct. This is the conceptual basis for the law.

39. Correct Answer: Both (A) and (R) are true, and (R) is the correct explanation of (A).

Assertion (A): Correct. If you double the amount of gas (moles), the space it occupies doubles, provided pressure and temperature don't change.

Reason (R): Correct. According to Avogadro's Law, volume is proportional to the number of moles (and thus molecules).

40. Correct Answer: (A) is false, but (R) is true

Assertion (A): Incorrect**.** The ideal gas equation ($PV = nRT$) is a theoretical model. Real gases deviate from this behavior because of intermolecular forces and the actual volume of gas particles.

Reason (R): Correct**.** Real gases only approximate ideal behavior when molecules are far apart (low pressure) and moving fast (high temperature).

MATRIX MATCHING TYPE

41. Correct Answer: a-Q, b-T, c-P,d-R

A. Joules (J) is the standard SI unit for energy
B. The volume occupied by 1 mole of any ideal gas at STP.
C. This equals 1 atm, the standard pressure reference.
D. Ergs are the unit of energy in the Centimetre-Gram-Second system.

42. Correct Answer: a-S, b-Q, c-R,d-P

A. Pressure and volume are inversely related.
B. Also known as the Pressure-Temperature law.
C. Volume increases as absolute temperature increases.
D. Volume is proportional to the number of moles (n).

COMPREHENSHION TYPE

43. Correct Answer: B) 4 atm

Explanation: According to Boyle's Law, $PV = k$ or $P = k\frac{1}{V}$ On a plot of P vs $\frac{1}{V}$, the slope is equal to the constant k (which is nRT).

Given: Slope (k) = 60 L.atm, Volume (V) = 15 L.

Calculation:

$P = \frac{slope}{V} = \frac{60}{15} = 4atm$

44. Correct Answer: B) 4 atm

Step 1 (Isothermal Compression): The gas is compressed from 6 L to 4 L at a constant temperature.

- $P_1 V_1 = P_2 V_2 \Rightarrow$ (2 atm X 6 L) = P_2 X4 L
- $P_2 = \frac{12}{4}$ = 3 atm

Explanation: The gas is then heated from $17^{\circ}C to 127^{\circ}C$ at a constant volume.

- Convert temperatures to Kelvin: T_2 = 17 + 273 = 290 K, T_3 = 127 + 273 = 400 K.
- $\frac{P_1}{T_1} = \frac{P_2}{T_2} \Rightarrow \frac{3}{290} = \frac{P_3}{400}$
- $P_3 = \frac{3 \times 400}{290} \approx 4.14 atm$

45. Correct Answer: A) 27.5 L

Explanation: Use the Combined Gas Law: $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

Initial State (1): P_1 = 6 atm, V_1 = 10 L, T_1 = 127 + 273 = 400 K.

Final State (2): P_2 = 3 atm (half of initial), T_2 = 277 + 273 = 550 K.

Calculation:

$V_2 = \frac{P_1 \times V_1 \times T_2}{P_2 \times T_1} = \frac{6 \times 10 \times 550}{3 \times 400}$

$V_2 = \frac{33000}{1200}$ = 27.5 L

46. Correct Answer: C) 273°C

Explanation: Identify variables:

- $V_1 = V$
- $V_2 = 2V$
- $T_1 = 0^{\circ}C$ = 273 K

Apply Charles's Law: $\frac{V}{273} = \frac{2V}{T_2}$

T_2 = 2 X 273 = 546K

Convert back to Celsius:

546 - 273 = 273^0 C

47. Correct Answer: D) 21.38 mL

Explanation: Identify variables:

- V_1 = 20 mL
- T_1 = 15 + 273 = 288 K
- T_2 = 35 + 273 = 308 K

Calculate V_2:

$V_2 = \frac{V_1 \times T_2}{T_1}$

$V_2 = \frac{20 \times 308}{288}$

$V_2 \approx$ 21.38mL

48. Correct Answer: B) double

Explanation:

Since volume is directly proportional to absolute temperature ($V \alpha T$), if you double the absolute temperature while keeping pressure constant, the volume must also double.

INTEGER TYPE:

49. Correct Answer: The weight of CO present in a 22.4 litres flask at STP is 28 g.

Explanation:

The volume **22.4 litres** is the standard molar volume of an ideal gas at **STP** (Standard Temperature and Pressure, which is $0^{\circ}C$ or $273.15\ K$ and $1 atm$).

- **Molar Volume:** $22.4\ L$ at STP contains $1 mole$ of gas.
- **Molar Mass of CO:** $C\ (12) + O\ (16) = 28\ g/mol$.
- **Calculation:** Since the flask contains 1 mole of CO, the weight is $28\ g$.

50. Correct Answer: The value of '100 - x' is approximately 6.46.

Explanation:

Find moles (n): Using the Ideal Gas Law PV = nRT (where R ≈ 0.0821 L. atm/K. mol):

$$\frac{PV}{RT} = \frac{16 \times 9}{0.0821 \times 300} = \frac{144}{24.63} \approx 5.8465$$

Find mass (x): Molar mass of CH_4 = 12 + (4 X 1) = 16g/mol.

x = n X Molar Mass} = 5.8465 X 16 ≈ 93.54g

Calculate 100 - x:

100 - 93.54 = 6.46

51. Correct Answer: The final pressure is 1000 mm Hg.

Explanation: **Apply Boyle's Law:** $P_1V_1 = P_2V_2$

600 X 500 = P_2 X 300

300,000 = 300 X P_2

$P_2 = \frac{300,000}{300}$ = 1000 mm Hg

52. Correct Answer: The volume will become 2 L at 600 K.

Explanation: Given: V_1 = 1L, T_1 = 300K, V_2 = 2L (at constant pressure).

- **Apply Charles's Law:** $\frac{V_1}{T_1} = \frac{V_2}{T_2}$
- $\frac{1}{300} = \frac{2}{T_2}$

T_2 = 2 X 300 = 600 K

Previous year questions

1. Correct Answer: D) Statement–I is not correct, but statement–II is correct

Explanation:

Gases at room temperature:

- H_2, N_2, O_2, F_2, Cl_2 → 5 diatomic gases
- He, Ne, Ar, Kr, Xe → 5 noble gases

Total = 10 gases

2. Correct Answer: C) High temperature and low pressure

Explanation:

Van der Waals equation becomes ideal gas equation when:

- **Volume of molecules is negligible**
- **Intermolecular attractions are negligible**

This happens at:

- **High temperature** → molecules move fast, attractions become insignificant
- **Low pressure** → molecules are far apart

3. Correct Answer:

Both statements are correct, but Statement–II is not the correct explanation of Statement–I

Explanation:

Both are **true**

But Statement–II does **not explain Statement–I**

- One is about **non-ideal solutions**
- Other is about **ideal solutions**

4. **Correct Answer: C) 1.00**

   Explanation: Pressure, $P = 16.4$ atm
   Temperature, $T = 200$ K
   Gas constant, $R = 0.082$ L atm $\text{mol}^{-1}\text{K}^{-1}$
   We are asked for concentration in mol L^{-1}, i.e. $\frac{n}{V}$
   Use Ideal Gas Equation:

   $$PV = nRT$$

   Rearrange:

   $$\frac{n}{V} = \frac{P}{RT}$$

   Substitute values:

   $$\frac{n}{V} = \frac{16.4}{0.082 \times 200}$$
   $$= \frac{16.4}{16.4}$$
   $$= 1.00 \text{ mol L}^{-1}$$

5. **Correct Answer: B) b, c, a only**

   Explanation: Compressibility factor:

   $$Z = \frac{PV}{RT}$$

   - Z = 1 → Ideal gas
   - Z > 1 → Positive deviation (repulsive forces dominate)
   - Z < 1 → Negative deviation (attractive forces dominate)

   From the graph:

   - Curves b, c, a always stay above Z = 1
   - Curves d, e, f dip below Z = 1 (negative deviation at some pressure)

6. **Correct Answer: A) $\frac{PM}{RT}$**

   Explanation:

   From ideal gas equation:

   $$\frac{n}{V} = \frac{P}{RT}$$

   So,

   $$\text{Density} = \frac{PM}{RT}$$

7. **Correct Answer: B) There is a complete disorder of molecules in gases**

   Explanation:

   Kinetic Theory of Gases says:

   - Gas molecules move randomly
   - They have no fixed positions
   - They are in a state of maximum disorder

8. **Correct Answer: A) CCl_4 + C_7H_8 (Toluene)**

   Explanation:

   - Both are non-polar
   - Similar molecular size
   - Similar intermolecular forces (London dispersion)
   - A–B ≈ A–A ≈ B–B
     Forms ideal solution

9. **Correct Answer: B) One**

   Explanation:

   Concept:

   - At low pressure (1 atm) and moderate temperature (273 K), gases behave almost ideally
   - For an ideal gas:

     $$Z = 1$$

10. **Correct Answer: C) (i), (ii), and (iii)**

    Explanation:

    Ideal Gas Assumptions:

    Gas molecules have negligible volume
    All collisions are perfectly elastic

    Distance between molecules is much larger than their size

7. Hydrogen and its Compounds

Solutions

Worksheet –1

1. **Correct Answer: B) Lavoisier; "Water maker"**

   Explanation: Antoine Lavoisier named hydrogen. The word comes from Greek: hydro = water, genes = maker, meaning "water former" because hydrogen forms water when it burns in oxygen.

2. **Correct Answer: C) Alkali metals and Halogens**

   Explanation: Hydrogen has one electron like alkali metals (can lose it to form H^+) and also needs one electron to complete its shell like halogens (can gain one to form H^-).

3. **Correct Answer: C) NaH**

   Explanation: In metal hydrides like sodium hydride (NaH), hydrogen gains an electron and exists as H^- (oxidation state –1).

4. **Correct Answer: B) Existing as diatomic molecules (X_2)**

   Explanation: Hydrogen (H_2) and halogens (F_2, Cl_2, etc.) exist as diatomic molecules, but alkali metals exist as single atoms in metallic form.

5. **Correct Answer: C) The fusion of hydrogen nuclei to form helium**

   Explanation: The sun and stars produce energy by nuclear fusion, where hydrogen nuclei combine to form helium, releasing huge amounts of energy.

6. **Correct Answer: B) Readily gives up its electron to reduce others .**

   Explanation: This is exactly what a reducing agent does.

7. **Correct Answer: B) H^+ ion by losing an electron**

   Explanation: Like alkali metals, hydrogen can lose one electron to form a unipositive ion (H^+).

8. **Correct Answer: B) CH_4 and CCl_4**

   Explanation: Both compounds show covalent bonding with carbon sharing electrons to form stable molecules.

9. **Correct Answer: B) They exist as diatomic molecules**

   Explanation: Elemental hydrogen exists as H_2, and halogens exist as X_2 (like F_2, Cl_2), so both are diatomic.

10. **Correct Answer: C) Metal hydride**

    Explanation: When hydrogen reacts with sodium, it forms sodium hydride (NaH), which is a metal hydride.

Worksheet-2

1. **Correct Answer: A) (Copper + dilute acid)**

   Explanation: Copper is less reactive than hydrogen, so it cannot displace hydrogen from acids. Other reactive metals can.

2. **Correct Answer: C) Mg**

   Explanation: Magnesium reacts with dilute HCl to give hydrogen:
   $Mg + 2HCl \rightarrow MgCl_2 + H_2\uparrow$

3. **Correct Answer: C) Hg**

   Explanation: Mercury is below hydrogen in the reactivity series, so it cannot displace hydrogen from acids.

4. **Correct Answer: A) Cu**

   Explanation: Copper does not react with dilute acids to produce hydrogen.

5. **Correct Answer: D) Producer gas**

   Explanation: Producer gas (mixture of CO and N_2) is used as a source for large-scale hydrogen production.

6. **Correct Answer: C) Trapped air**
   Explanation: The apparatus initially contains air, which must be flushed out to avoid an explosive mixture.
7. **Correct Answer: C) To avoid hydrogen escaping through the funnel**
   Explanation: If the stem is above the liquid level, hydrogen can escape through the funnel instead of the delivery tube.
8. **Correct Answer: A) Rapid oxidation of hydrogen in air**
   Explanation: Hydrogen burns with oxygen, producing water and a characteristic "pop" sound.
9. **Correct Answer: B) Keeping away from open flames**
   Explanation: Hydrogen is highly flammable and can ignite easily near flames or sparks.
10. **Correct Answer: D) Pt and NaOH**
    Explanation: Nascent hydrogen is generated during certain chemical reactions, such as when metals react in alkaline medium with catalysts like platinum.

Worksheet-3

1. **Correct Answer: B) Oxidation**

   Explanation: Oxidation is defined as the loss of electrons.

2. **Correct Answer: A) Zn**

   Explanation:

   - Zinc **loses electrons** and gets oxidized:
     $Zn \rightarrow Zn^{2+} + 2e^-$
   - Since zinc gives electrons to Cu^{2+}, it **reduces Cu^{2+} to Cu**
     So, **Zn is the reducing agent**.

3. **Correct Answer: C) Chlorine**

   Explanation:
   Chlorine (Cl_2) is a gas and a strong oxidizing agent because it gains electrons easily and forms Cl^- ions.

4. **Correct Answer: B) MnO_4^- is reduced to Mn^{2+}**

   Explanation:
   Potassium permanganate (purple) gets reduced in acidic medium to Mn^{2+}, which is colorless. That's why the purple color disappears.

5. **Correct Answer: C) Rusting of iron**
   Explanation:
   Rusting happens slowly over time and involves oxidation of iron and reduction of oxygen, making it a slow redox process.
6. **Correct Answer: C) Protons**
   Explanation:
   Isotopes of an element always have the **same number of protons** (atomic number) but different numbers of neutrons.
   Hydrogen isotopes:
   - Protium: 1p, 0n
   - Deuterium: 1p, 1n
   - Tritium: 1p, 2n
7. **Correct Answer: C) Occur simultaneously**
   Explanation:
   You cannot have oxidation without reduction and vice versa. When one substance loses electrons, another must gain them.
8. **Correct Answer: B) Orange to green**
   Explanation:
   Acidified $K_2Cr_2O_7$ (orange) is reduced to Cr^{3+} (green) in the presence of a reducing agent.
9. **Correct Answer: B) Deuterium**

   Explanation:

   - Protium: 1 proton, 0 neutrons
   - **Deuterium: 1 proton, 1 neutron**
   - Tritium: 1 proton, 2 neutrons

10. **Correct Answer: B) Blue color with starch**

    Explanation:
    A strong oxidizing agent oxidizes iodide ions (I^-) to iodine (I_2).
    Iodine forms a blue-black complex with starch, confirming its presence.

Worksheet-4

1. **Correct Answer: B) Bent**
   Explanation:
   Oxygen has two bonding pairs and two lone pairs of electrons. Lone pair–bond pair repulsion bends the molecule, giving a **V-shaped (bent)** structure with an angle of about **104.5°**.
2. **Correct Answer: C) sp^3**
   Explanation:
   Oxygen forms four electron domains (2 bond pairs + 2 lone pairs), so it uses **sp^3** hybrid orbitals.
3. **Correct Answer: B) 1.84 Debye**
   Explanation:
   Water is a polar molecule due to the bent shape and difference in electronegativity between O and H, giving a dipole moment of about 1.84 D.
4. **Correct Answer: B) Ice has a tetrahedral open cage structure**
   Explanation:
   In ice, hydrogen bonding forms an open hexagonal/tetrahedral lattice, increasing volume and lowering density compared to liquid water, so ice floats.
5. **Correct Answer: C) It dissolves many polar and ionic substances**
   Explanation:
   Due to its polarity and hydrogen bonding, water dissolves salts, acids, bases, and many polar compounds.
6. **Correct Answer: C) Both an acid and a base**
   Explanation:
   Water is amphoteric.
   - As an acid: $H_2O \rightarrow H^+ + OH^-$
   - As a base: $H_2O + H^+ \rightarrow H_3O^+$
7. **Correct Answer: A) $Mg + 2H_2O \rightarrow Mg(OH)_2 + H_2$**
   Explanation:
   A reducing agent loses electrons and causes another substance to be reduced. Here, water is reduced to hydrogen gas (H_2), while magnesium is oxidized. Water donates hydrogen, helping Mg get oxidized—so water behaves as a reducing agent in this reaction.
8. **Correct Answer: A) $Na_2CO_3 \cdot 10H_2O$**
   Explanation:
   Efflorescent substances lose water of crystallization when exposed to air. Washing soda ($Na_2CO_3 \cdot 10H_2O$) loses some of its water and becomes powdery.
9. **Correct Answer: B) Increases**
   Explanation:
   KNO_3 is a solid whose solubility increases sharply with temperature, which is why it's often used in crystallization experiments.
10. **Correct Answer: B) They absorb moisture from the air**
    Explanation:
    Hygroscopic substances absorb water vapor from the atmosphere but do not necessarily dissolve in it (e.g., silica gel, concentrated H_2SO_4).

Worksheet-5

1. **Correct Answer: C) Distilled water**
   Explanation:
   Distilled water is soft water and does not contain calcium or magnesium ions, so soap forms lather easily.
2. **Correct Answer: A) Potash alum**
   Explanation:
   Potash alum ($K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$) is commonly used as a coagulant to remove suspended impurities from water.
3. **Correct Answer: C) Is a sodium aluminium silicate**
   Explanation:
   Zeolites (permutit) are sodium aluminium silicates that exchange Na^+ ions with Ca^{2+} and Mg^{2+} ions in hard water.
4. **Correct Answer: A) Bicarbonates of Ca**
   Explanation:
   Temporary hardness is due to calcium and magnesium bicarbonates, which can be removed by boiling.

5. **Correct Answer: B) It contains dissolved calcium and magnesium salts**
Explanation:
Ca^{2+} and Mg^{2+} ions react with soap to form **scum**, preventing lather formation.
6. **Correct Answer: B) Boiling**
Explanation:
Boiling decomposes bicarbonates into insoluble carbonates, which precipitate out.
7. **Correct Answer: C) Boiling**
Explanation:
Permanent hardness is caused by chlorides and sulphates of Ca and Mg, which are not removed by boiling.
8. **Correct Answer: C) Forms scale deposits**
Explanation:
Scale reduces heat transfer efficiency and can damage the boiler.
9. **Correct Answer: C) Hydrogen ions**
Explanation:
Cation exchange resins replace Ca^{2+} and Mg^{2+} ions with H^+ ions, softening the water.
10. **Correct Answer: B) Sodium chloride**
Explanation:
Common salt (NaCl) restores Na^+ ions in the zeolite/permutit so it can be reused.

Competitive Corner

SINGLE CORRECT OPTION TYPE

1. **Correct Answer: C) Low conductance**

Explanation: Pure (distilled) water contains only neutral molecules and is almost completely free of ions like sodium or chloride that are needed to carry an electric charge. While it has a very tiny amount of ions due to self-ionization, its conductivity is extremely low, ranging from 0.5 to 3 µS/cm

2. **Correct Answer: C) rain water**

Explanation: Rainwater is considered the purest form of natural water because it is formed through the natural process of distillation—evaporation followed by condensation—which leaves most terrestrial impurities behind.

3. **Correct Answer: D) V-shape**

Explanation: According to VSEPR theory, the oxygen atom has two bonding pairs and two lone pairs of electrons. These lone pairs repel the bonding pairs, pushing the hydrogen atoms closer together and creating a **bent** or **V-shape**

4. **Correct Answer: B)** 104.5°

Explanation: In a perfect tetrahedron, the angle would be 109.5° However, the strong repulsion from the two lone pairs on the oxygen atom compresses the angle between the hydrogen atoms down to approximately 104.5°.

5. **Correct Answer: A) 1.84 D**

Explanation: Due to its bent shape and the high electronegativity of oxygen compared to hydrogen, water is a highly polar molecule. The net dipole moment resulting from these individual bond polarities is **1.84 Debye**.

6. **Correct Answer: A) Covalent bond**

Explanation: Within a single water molecule, the oxygen and hydrogen atoms stay together by **sharing electrons**, which is the definition of a covalent bond.

7. **Correct Answer: D) hydrogen bonds**

Explanation: While covalent bonds hold the atoms *inside* a molecule together, **hydrogen bonds** are the attractive forces *between* different water molecules. The slightly positive hydrogen of one molecule is attracted to the slightly negative oxygen of a neighbour.

8. **Correct Answer: D) 4**

Explanation: Each water molecule can form four hydrogen bonds: two through its two hydrogen atoms (acting as donors) and two through the two lone pairs on its oxygen atom (acting as acceptors).

9. **Correct Answer: B) ice crystals have hollow hexagonal arranged of H_2O molecules**
   Explanation: In its solid state (ice), water molecules form a rigid, hexagonal lattice held by hydrogen bonds. This structure is very "open" and contains hollow spaces, making ice less dense (lighter) than liquid water, where molecules can pack more closely together.
10. **Correct Answer: C) Saturation of water with $CaSO_4$**
    Explanation: Hardness in water is caused by dissolved salts of Calcium and Magnesium. While CaCO3 (Calcium Carbonate) and MgCO3 (Magnesium Carbonate) are compounds of these metals, they are insoluble in water and will not dissolve to create "hardness" unless CO_2 is present. $CaSO_4$ (Calcium Sulfate) is soluble enough to release Ca^{2+} ions into the water, resulting in permanent hardness.
11. **Correct Answer: C) Ca^{2+} and Mg^{2+}**
    Explanation**:** By definition, water hardness is the measure of the concentration of multivalent cations. The most common ions that react with soap to form "scum" and cause scaling in pipes are Calcium (Ca^{2+}) and Magnesium (Mg^{2+}.)
12. **Correct Answer: B) $Mg(HCO_3)_2$**
    Explanation: Hardness is categorized into two types:
    - Temporary: Caused by dissolved bicarbonates (like $Mg(HCO_3)_2$ or $Ca(HCO_3)_2$).
    - Permanent: Caused by chlorides and sulfates (like $MgSO_4$ or $CaCl_2$).

    Since the question asks for temporary hardness, the bicarbonate option is correct.
13. **Correct Answer: D) Temporary hardness only**
    - Explanation: Boiling decomposes dissolved bicarbonates into insoluble carbonates, which precipitate out as solids.

      e.g., $Ca(HCO_3)_2 \xrightarrow{\Delta} CaCO_3 \downarrow + H_2O + CO_2$

      Boiling cannot remove permanent hardness (sulfates and chlorides), as those salts remain dissolved even at boiling temperatures.
14. **Correct Answer: B) Ca^{2+} ions**
    Explanation: Zeolite (Permutit process) acts as an ion exchanger. The sodium ions (Na^+) in the zeolite structure are loosely held and are replaced by the "hardness" ions (Ca^{2+} or Mg^{2+}) from the water as it passes through. This softens the water by removing the problematic minerals.
15. **Correct Answer: B) Slow cooling of solution**
    Explanation:
    Crystallization involves forming solid crystals from a hot saturated solution. When the solution cools slowly, the solute molecules have time to arrange themselves into a well-ordered crystal lattice. Rapid cooling causes irregular, small crystals or amorphous solids to form. Hence, slow cooling produces good-quality crystals.
16. **Correct Answer: D) Electrolysis of sodium chloride solution**
    Explanation:
    Electrolysis of NaCl gives H_2 and Cl_2, not water.
17. **Correct Answer: B) Silver nitrate solution**
    Explanation:
    $AgNO_3$ reacts with AsH_3 and PH_3, removing them as solid products.
18. **Correct Answer: C) Lead sulfide (PbS)**
    Explanation:
    $Pb^{2+} + H_2S \rightarrow PbS$ (black ppt) $+ 2H^+$
19. **Correct Answer: B) Carbon dioxide and sulfur dioxide**
    Explanation:
    KOH absorbs acidic gases like CO_2 and SO_2.
20. **Correct Answer: B) To remove moisture**

    Explanation:
    $CaCl_2$ is a **drying agent** that absorbs water vapor.

21. **Correct Answer: B) To detect gas leakage**
Explanation:
Leaking gas forms bubbles in soapy water.
22. **Correct Answer: B) 277 K (4°C)**
Explanation:
Water is densest at 4°C (277 K) due to hydrogen bonding behavior.
23. **Correct Answer: B) 18 g**
Explanation: Step 1: Given mass of hydrogen
    - $H_2 = 2\ g$
    - Molar mass of $H_2 = 2\ g/mol$

Moles of $H_2 = \frac{2}{2} = 1$ mole
Step 2: Use mole ratio
From the equation:
    - $2\text{moles}\ H_2 \rightarrow 2\text{moles}\ H_2O$

So,
    - $1\text{mole}\ H_2 \rightarrow 1\text{mole}\ H_2O$
    - Step 3: Convert to mass
    - Molar mass of $H_2O = 18\ g/mol$

Mass of water formed = **1 × 18 = 18 g**
24. **Correct Answer: B)**
Explanation: B) $NaOH + H_2O \rightarrow NaOH(aq) + H_2$
This reaction is incorrect
    - Sodium hydroxide does NOT react with water to produce hydrogen
    - So this is NOT a method

ONE OR MORE THAN ONE CORRECT OPTION TYPE

25. **Correct Answer: B, C, and D**
Explanation**:** Permanent hardness is caused by chlorides and sulfates of Calcium and Magnesium.
    - B (Ion-exchange): Replaces Ca^{2+}/Mg^{2+} with Na^+ or H^+.
    - C (Calgon's): Uses sodium hexametaphosphate to sequester the hardness-causing ions.
    - D (Washing soda): Precipitates the ions as insoluble carbonates ($M^{2+} + Na_2CO_3 \rightarrow MCO_3\downarrow + 2Na^+$).

26. **Correct Answer: A, B, and C**
Explanation: These are amphoteric metals (or metals that react with base). They react with strong alkalis like sodium hydroxide to form soluble complexes and release hydrogen gas.
    - Zn: $Zn + 2NaOH \longrightarrow Na_2ZnO_2$ (Sodium zincate) + $H_2\uparrow$
    - Sn: $Sn + 2NaOH + H_2O$ Na_2SnO_3 (Sodium stannate) + $2H_2\uparrow$
    - Al: $2Al + 2NaOH + 6H_2O$ $2Na[Al(OH)_4]$ (Sodium aluminate) + $3H_2\uparrow$
    - *Note: Copper (Cu) does not react with NaOH to liberate hydrogen.*

27. **Correct Answer: A and B**
Explanation: Hydrogen gas can reduce the oxides of metals that are less reactive than hydrogen in the reactivity series.
    - Cupric oxide (CuO): $CuO + H_2 \longrightarrow Cu + H_2O$ (Occurs easily)
    - Ferric oxide (Fe_2O_3): $Fe_2O_3 + 3H_2 \longrightarrow 2Fe + 3H_2O$ (Occurs at high temperatures)
    - Zinc oxide and Sodium oxide**:** These metals (Zn, Na) are more reactive than hydrogen, so their oxides are too stable to be reduced by H_2 gas under standard laboratory conditions.

28. **Correct Answer: A and C**
Explanation:
    - **A:** Iron is above hydrogen in the reactivity series and displaces it from dilute acids: $Fe + H_2SO_4 \longrightarrow FeSO_4 + H_2\uparrow$
    - **C:** Sodium reacts vigorously with the acidic hydrogen in the hydroxyl group of ethanol: $2C_2H_5OH + 2Na \longrightarrow 2C_2H_5ONa$ (Sodium ethoxide) + $H_2\uparrow$

29. **Correct Answer: A, B, and D**

Explanation:
An oxidizing agent is a substance that gains electrons (it gets reduced) and causes another substance to be oxidized.

- **$KMnO_4$ (Potassium permanganate)** → Strong oxidizing agent (Mn^{7+} gets reduced)
- **H_2O_2 (Hydrogen peroxide)** → Can act as an oxidizing agent (gets reduced to H_2O)
- **O_2 (Oxygen)** → Classic oxidizing agent (accepts electrons during combustion and oxidation)
- **CO (Carbon monoxide)** → Not an oxidizing agent; it is a **reducing agent** (it reduces metal oxides to metals)

30. **Correct Answer: A, B, and C**

Explanation:
A reducing agent is a substance that loses electrons and causes another substance to be reduced.

- **Sodium (Na)** → Loses electrons easily → strong reducing agent
- **Carbon (C)** → Reduces metal oxides to metals (e.g., in blast furnace)
- **Carbon monoxide (CO)** → Reduces metal oxides (e.g., Fe_2O_3 → Fe)
- **Chlorine (Cl_2)** → Gains electrons → acts as an **oxidizing agent**, not reducing

ASSERTION AND REASON TYPE

31. **Correct Answer: Both A and R are true, and R is the correct explanation of A.**

Explanation:
Zinc reacts with dilute HCl to produce hydrogen gas:
$Zn + 2HCl \rightarrow ZnCl_2 + H_2\uparrow$
Zinc being more reactive than hydrogen displaces it from the acid.

32. **Correct Answer: A**

Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

Explanation:
Assertion (A): Silver nitrate solution is used in hydrogen purification to remove arsine and phosphine.

- True. In hydrogen purification, impurities like **arsine (AsH_3)** and **phosphine (PH_3)** are removed because they can poison catalysts in reactions where pure hydrogen is needed.

Reason (R): Arsine and phosphine are oxidized by silver nitrate to form silver metal and corresponding acids.

- True. The chemical reactions are:

$AsH_3 + 6AgNO_3 + 3H_2O \rightarrow 6Ag + H_3AsO_3 + 6HNO_3$
$PH_3 + 8AgNO_3 + 4H_2O \rightarrow 8Ag + H_3PO_3 + 8HNO_3$

33. **Correct Answer:Both A and R are true, and R is the correct explanation of A.**

Explanation:
Permanent hardness is due to $CaCl_2$, $MgCl_2$, $CaSO_4$, $MgSO_4$, which do not precipitate on boiling, so boiling cannot remove it.

34. **Correct Answer: Both A and R are true, and R is the correct explanation of A.**

Explanation:
Na_2CO_3 reacts as follows:
$CaSO_4 + Na_2CO_3 \rightarrow CaCO_3\downarrow + Na_2SO_4$
$MgCl_2 + Na_2CO_3 \rightarrow MgCO_3\downarrow + 2\ NaCl$
The insoluble carbonates precipitate, removing hardness.

35. **Correct Answer:**

A: Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

Explanation:
This reaction is highly exothermic, and the rapid burning of hydrogen produces the characteristic "pop" sound.

MATRIX MATCHING TYPE

36. Correct Answer: a ⟶Q P, Q, R, S; b ⟶Q P; c ⟶Q R, S; d ⟶Q Q, R, S

A. Hard water is defined by the presence of dissolved minerals, specifically calcium (Ca^{2+}) and magnesium (Mg^{2+}) ions. These ions can be present in the form of bicarbonates (like $Ca(HCO_3)_2$) or sulfates (like $MgSO_4$).

B. Temporary hardness is specifically caused by the presence of dissolved bicarbonates of calcium and magnesium, such as $Ca(HCO_3)_2$. It is called "temporary" because it can be removed simply by boiling.

C. In the zeolite (permutit) process, hard water passes through a bed of hydrated sodium alumino-silicate. The sodium ions (Na^+) in the zeolite are exchanged for the Ca^{2+} and Mg^{2+} ions in the hard water to soften it.

D. Permanent hardness is caused by chlorides and sulfates of calcium and magnesium, such as $MgSO_4$. Because it is fundamentally due to the presence of Ca^{2+} and Mg^{2+} ions, these options also apply.

37. Correct Answer: a ⟶Q S; b ⟶Q P; c ⟶Q; d ⟶Q R

A. Protium ($_1^1H$) is the most common isotope. It consists of 1 Proton (1P) and 1 Electron ($1e^-$), but has no neutrons (0n).

B. Deuterium ($_1^2H$ or D) contains 1 Proton (1P), 1 Electron ($1e^-$), and 1 Neutron (1n)

C. Tritium ($_1^3H$ or T) is the radioactive isotope. It consists of 1 Proton (1P), 1 Electron ($1e^-$), and 2 Neutrons (2n).

D. Dihydrogen (H_2) is a molecule formed by two hydrogen atoms. Since each hydrogen atom (protium) has 1 proton and 1 electron, the H_2 molecule has a total of 2 Protons (2P) and 2 Electrons ($2e^-$).

COMPREHENSHION TYPE

38. Correct Answer: B) Lose an electron

Explanation: Alkali metals have one electron in their outermost shell and usually lose it to form positive ions.

Hydrogen also has one electron and can lose it to form H^+.

39. Correct Answer: B) Diatomic molecular form

Explanation:

Halogens exist as diatomic molecules (Cl_2, F_2, etc.) and so does hydrogen (H_2).

Hydrogen also needs one electron to complete its shell, like halogens.

40. Correct Answer: C) Either gain or lose one electron

Explanation:

Hydrogen can lose one electron to form H^+ (ionic) or gain one electron to form H^- (ionic) or share one electron to form covalent bonds.

41. Correct Answer: C) $Ba(OH)_2$ Solution

Explanation: High purity hydrogen (>99.9%) is obtained in the laboratory by the electrolysis of a warm aqueous solution of barium hydroxide using nickel electrodes.

42. Correct Answer: B) Fe

Explanation**:** As shown in the provided passage (iii), Iron (Fe) reacts with steam to form magnetic oxide of iron (Fe_3O_4) and hydrogen gas. Mg can also react with steam, but Fe is the classic example for "passing steam" over the metal.

43. Correct Answer:Na (and K)

Explanation: According to passage (i), highly active alkali metals like Sodium (Na) and Potassium (K) react vigorously with water even at room temperature

INTEGER TYPE:

44. Correct Answer:

The atomic number of hydrogen is 1.

45. Correct Answer::

In the Earth's crust, hydrogen is present mainly in water. There are 2 hydrogen atoms in a single water molecule (H_2O).

46. Correct Answer::

Hydrogen forms diatomic molecules. The number of atoms in one hydrogen molecule (H_2) is 2.

47. Correct Answer::

Hydrogen is the first (or most) abundant element in the universe by mass, making up approximately 75% of all normal matter.

Previous year questions

1. Correct Answer:D) $Mg(OH)_2$, $CaCO_3$

Explanation**:** When water containing bicarbonates is boiled, they decompose. Calcium bicarbonate forms insoluble calcium carbonate. However, magnesium bicarbonate forms magnesium hydroxide because the solubility product (K_{sp}) of $Mg(OH)_2$ is much lower than that of $MgCO_3$, making the hydroxide precipitate more readily.

2. Correct Answer:A) placing in wax lined plastic bottle and kept in dark

Explanation**:** H_2O_2 is highly unstable and decomposes when exposed to light or rough surfaces like glass.

- Light**:** Photochemical decomposition occurs, so it must be kept in the dark.
- Container**:** Glass containers have alkali traces that catalyze decomposition; therefore, wax-lined plastic or urea-stabilized bottles are used.

3. Correct Answer:A) Reducing agent, Oxidizing agent

Explanation:

- With KIO4**:** H_2O_2 reduces Periodate (IO_4^-) to Iodate (IO_3^-). In this reaction, H_2O_2 is a reducing agent.
- With NH_2OH: H_2O_2 oxidizes Hydroxylamine (NH_2OH) to compounds like nitrous acid or nitrogen. Here, H_2O_2 acts as an oxidizing agent.

4. Correct Answer:B) Calgon's method

Explanation: Sodium hexametaphosphate ($Na_6P_6O_{18}$) is commercially known as Calgon. It works by sequestering Ca^{2+} and Mg^{2+} ions into a soluble complex, preventing them from reacting with soap to form scum.

5. Correct Answer:A) 111.5° **,** 90.2°

Explanation: H_2O_2 has a non-planar "open book" structure.

- In the gas phase, the dihedral angle is larger (111.5°) due to the absence of intermolecular forces.
- In the solid phase, strong hydrogen bonding between molecules compresses the structure, reducing the angle to 90.2° .

6. Correct Answer:A) 3.8 K

Explanation: * The temperature of maximum density for H_2O is $3.98^\circ C$ (277.13 K).

- The temperature of maximum density for D_2O (Heavy Water) is $11.2^\circ C$ (284.35 K).
- The difference is approximately $7.2^\circ C$. Note: While the arithmetic difference between

the provided options often points to the physical properties of the isotopes,

7. Correct Answer: B) $Mg(HCO_3)_2$ $Mg(OH)_2$

Explanation**:** while calcium precipitates as a carbonate, magnesium uniquely precipitates as a hydroxide because $Mg(OH)_2$ is more insoluble than $MgCO_3$ at boiling temperatures.

8. Correct Answer: D

Hydrogen peroxide (H2O2) has a non-planar, open-book structure. The dimensions change depending on whether it is in the gas phase or the solid (crystalline) phase due to intermolecular forces like hydrogen bonding.

- Gas Phase**:** Intermolecular forces are negligible. The dihedral angle is larger to minimize lone pair-lone pair repulsions. The angle is $11.5^{\circ}C$
- Solid Phase**:** Hydrogen bonding between molecules causes the structure to contract. The dihedral angle decreases significantly to $90.2^{\circ}C$

9. Correct Answer: C

Calgon" is a trade name for Sodium hexametaphosphate ($Na_6P_6O_{18}$).

- The Process**:** When Calgon is added to hard water, it reacts with the Ca^{2+}} and Mg^{2+} ions (which cause hardness).
- Mechanism: It does not precipitate these ions. Instead, it "sequesters" them by forming a soluble complex. This prevents the calcium and magnesium ions from reacting with soap

10. Correct Answer: C

NH_3 (Ammonia): Electron-rich. Nitrogen (Group 15) has 5 valence electrons. It uses 3 for bonding with Hydrogen and has one lone pair remaining.